THREE DIMENSIONAL DYNAMICS OF RIGID BODIES

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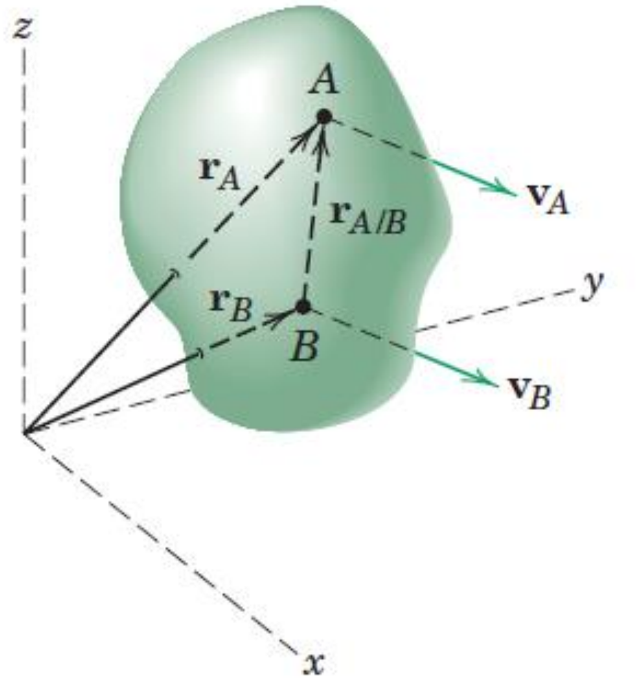
Translation

- Rectilinear Translation
- Curvilinear Translation

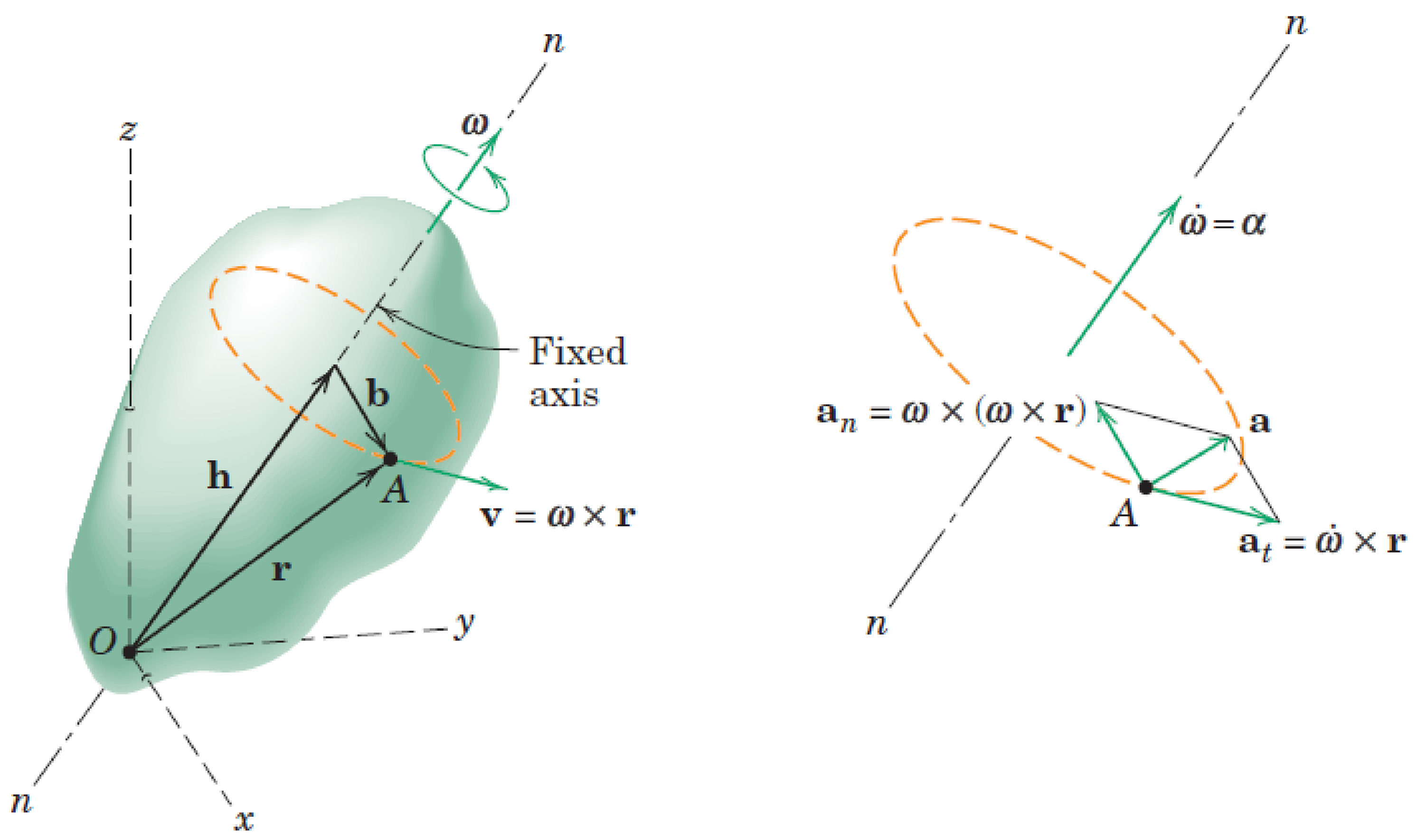
$$r_A = r_B + r_{A/B}$$

$$v_A = v_B$$

$$a_A = a_B$$



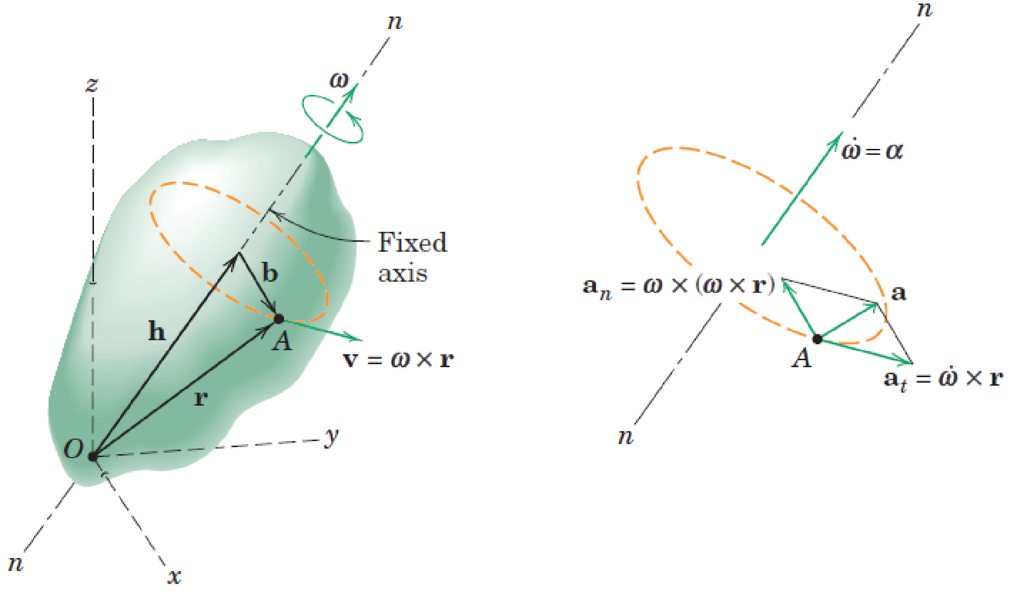
Fixed Axis Rotation



Fixed Axis Rotation

$$\mathrm{v} = \omega \times \mathrm{r}$$

$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$



$$\mathrm{r} = \mathrm{h} + \mathrm{b}$$

$$\mathrm{v} = \omega \times (\mathrm{h} + \mathrm{b}) = \omega \times \mathrm{b}$$

$$a_n = \omega \times (\omega \times r) = b\omega^2 \qquad a_t = \dot{\omega} \times r = b\alpha$$

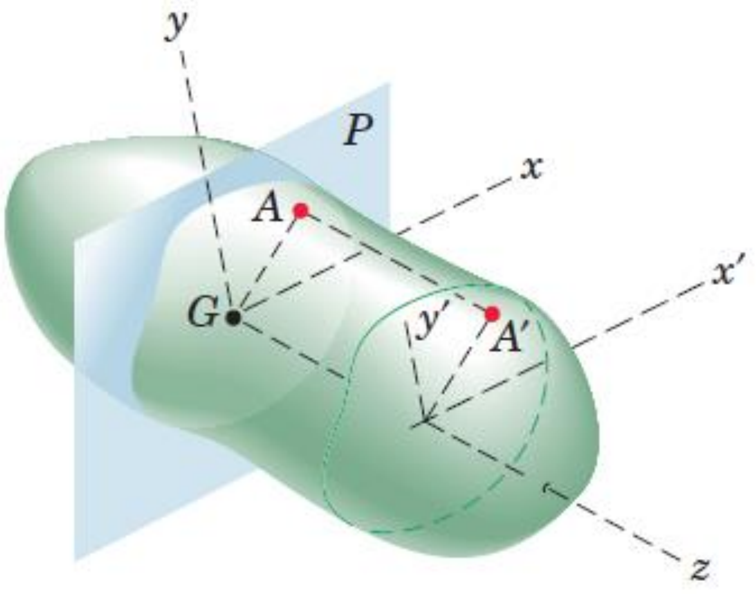
$$\mathrm{v}.\omega = 0 \qquad \mathrm{v}.\dot{\omega} = 0 \qquad a.\omega = 0 \qquad a.\dot{\omega} = 0$$

Parallel Plane Motion

Plane Motion: When all points in a rigid body move in planes which are parallel to a fixed plane *P*.

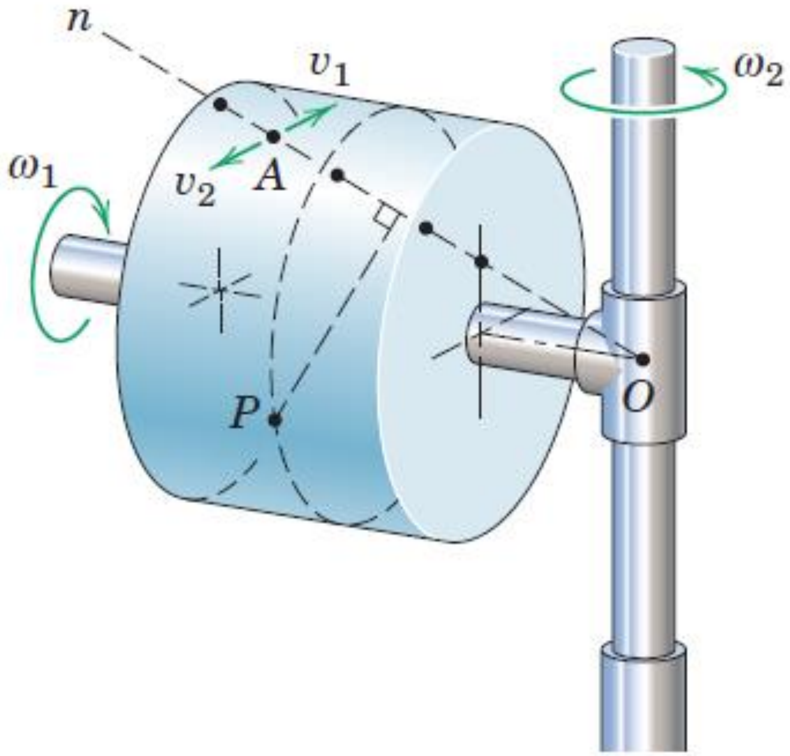
Plane of Motion: The reference plane is customarily taken through the mass center *G* and is called the <u>*plane of motion*</u>.

Because each point in the body, such as *A′*, has a motion identical with the motion of the corresponding point (*A*) in plane *P*, it follows that the kinematics of plane motion provides a complete description of the motion when applied to the reference plane.



Instantaneous axes of Rotation

- *A solid cylindrical rotor made of clear plastic containing many black particles embedded in the plastic*
- *The rotor is spinning about its shaft axis at the steady rate* ω_1 *, and its shaft, in turn, is rotating about the fixed vertical axis at the steady rate* ω_2 *, with rotations in the directions indicated*
- *If the rotor is photographed at a certain instant during its motion, the resulting picture would show one line of black dots sharply defined, indicating that, momentarily, their velocity was zero*
- *This line of points with no velocity establishes the instantaneous position of the axis of rotation O-n.*
- *Any dot on this line, such as A, would have equal and opposite velocity components,* v_1 *due to* ω_1 *and* v_2 *due to* ω_2*.*

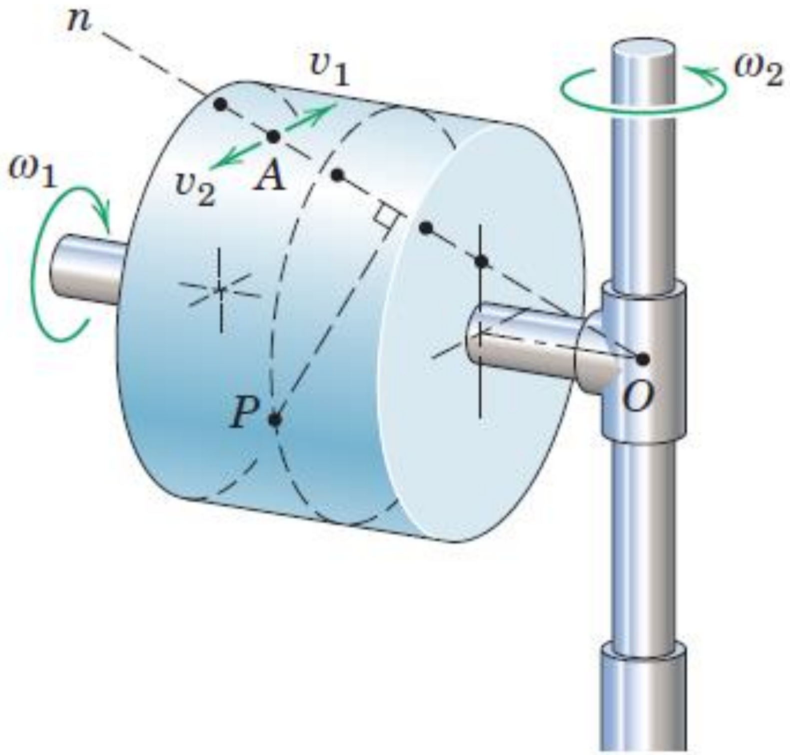


Instantaneous axes of Rotation

- *All other dots such as the one at P, would appear blurred, and their movements would show as short streaks in the form of small circular arcs in planes normal to the axis O-n. Thus, all particles of the body, except those on line O-n, are momentarily rotating in circular arcs about the instantaneous axis of rotation*

- *If a succession of photographs were taken, we would observe in each photograph that the rotation axis would be defined by a new series of sharply-defined dots and that the axis would change position both in space and relative to the body.*

- *For rotation of a rigid body about a fixed point, then, it is seen that the rotation axis is, in general, not a line fixed in the body.*

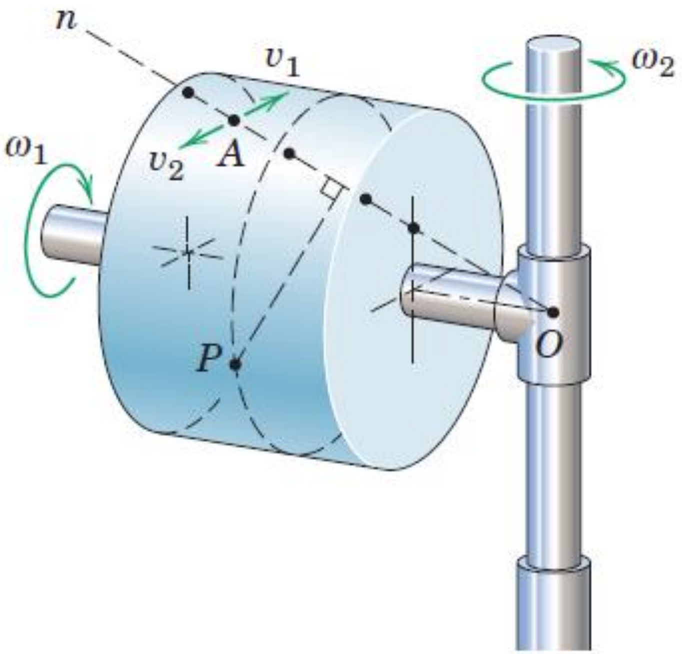


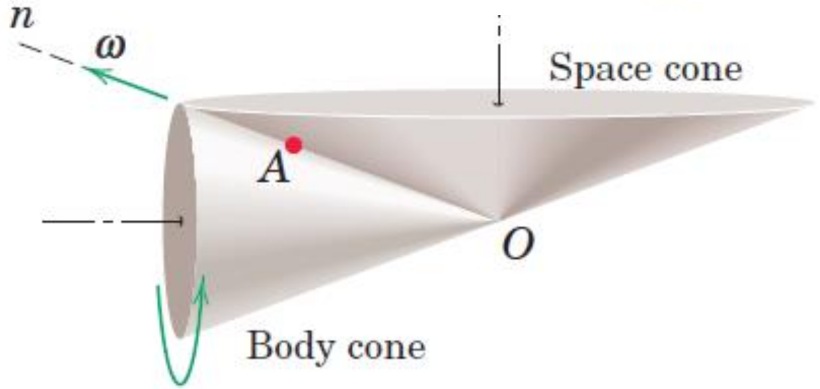
Body and Space Cones

- Relative to the plastic cylinder, the instantaneous axis of rotation *O-A-n* generates a right-circular cone about the cylinder axis called the *body cone*.

- As the two rotations continue and the cylinder swings around the vertical axis, the instantaneous axis of rotation also generates a right-circular cone about the vertical axis called the *space cone*

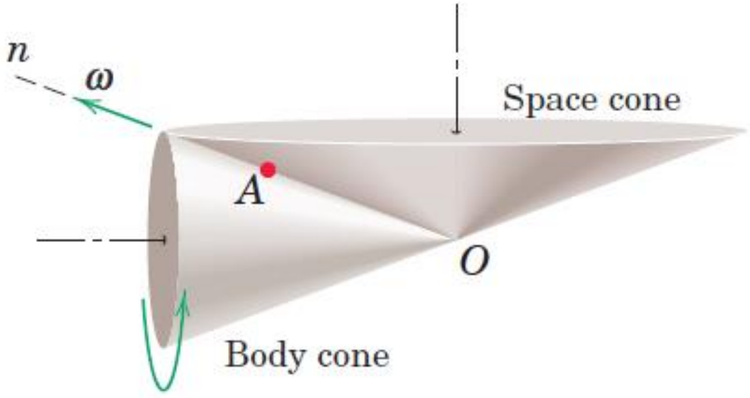
- These cones are shown in figure for this particular example.



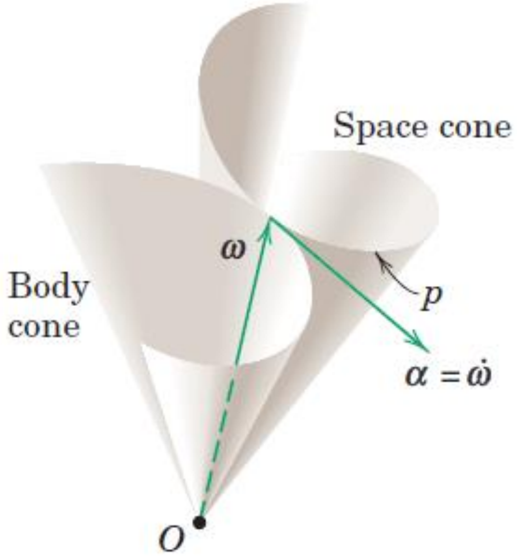


Body and Space Cones

- Body cone rolls on the space cone and that the angular velocity ω of the body is a vector which lies along the common element of the two cones



- For a more general case where the rotations are not steady, the space and body cones are not right-circular cones but the body cone still rolls on the space cone.

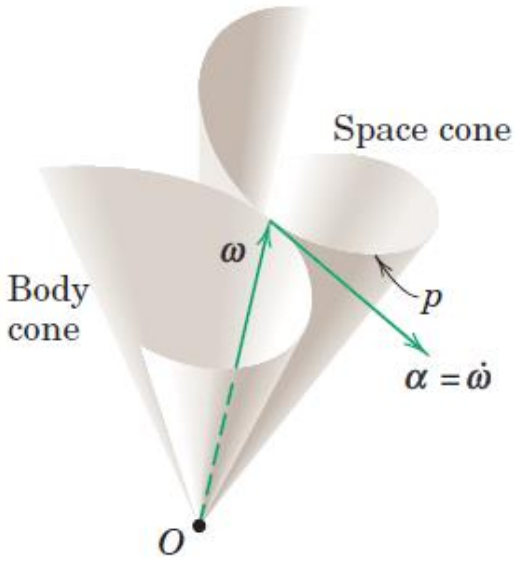


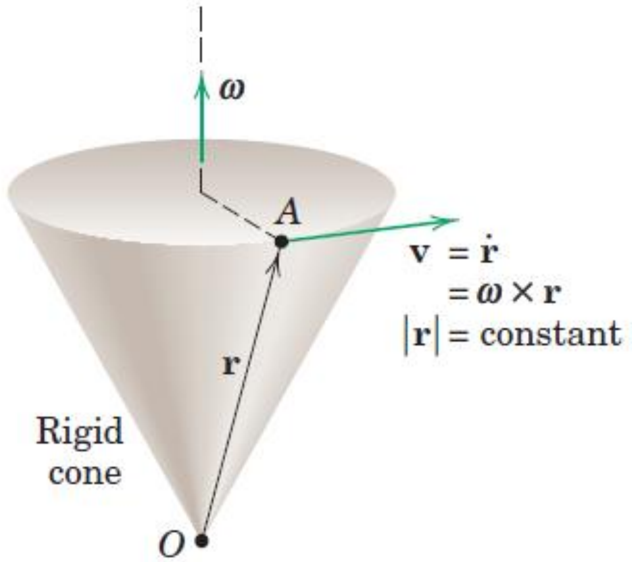
Angular Acceleration

- The angular acceleration $\boldsymbol{\alpha}$ of a rigid body in three-dimensional motion is the time derivative of its angular velocity, $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$

- In contrast to the case of rotation in a single plane where the scalar α measures only the change in magnitude of the angular velocity, in three-dimensional motion the vector $\boldsymbol{\alpha}$ reflects the change in direction of $\boldsymbol{\omega}$ as well as its change in magnitude.

- Thus in Figure where the tip of the angular velocity vector $\boldsymbol{\omega}$ follows the space curve p and changes in both magnitude and direction, the angular acceleration $\boldsymbol{\alpha}$ becomes a vector tangent to this curve in the direction of the change in $\boldsymbol{\omega}$.



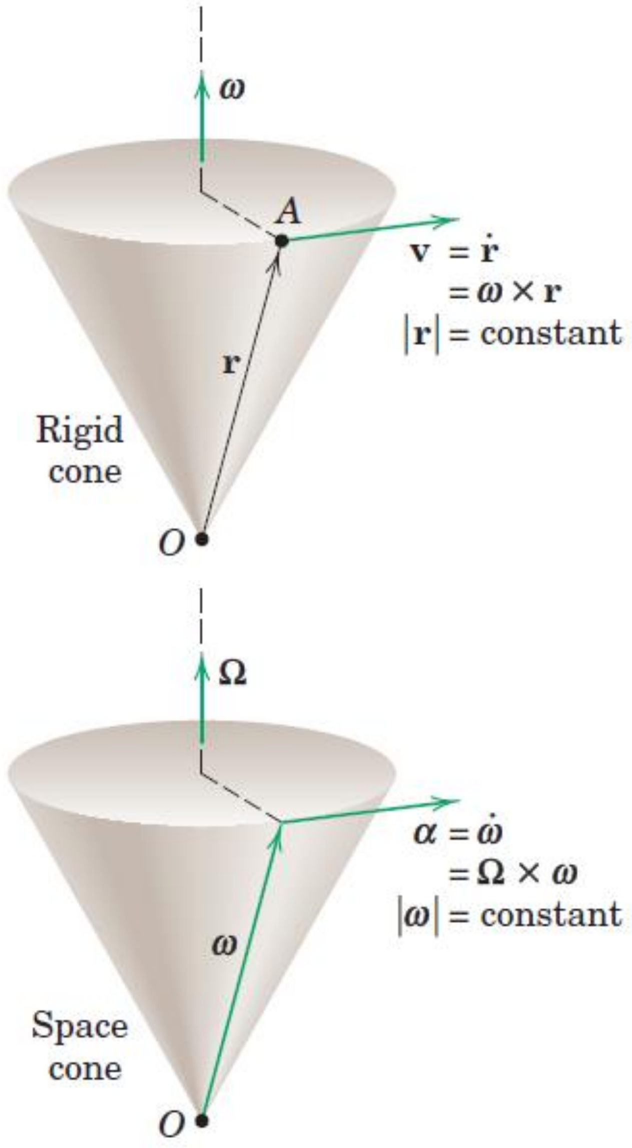


Angular Acceleration

- When the magnitude of $\boldsymbol{\omega}$ remains constant, the angular acceleration $\boldsymbol{\alpha}$ is normal to $\boldsymbol{\omega}$. For this case, if we let $\boldsymbol{\Omega}$ stand for the angular velocity with which the vector itself rotates (precesses) as it forms the space cone, the angular acceleration may be written

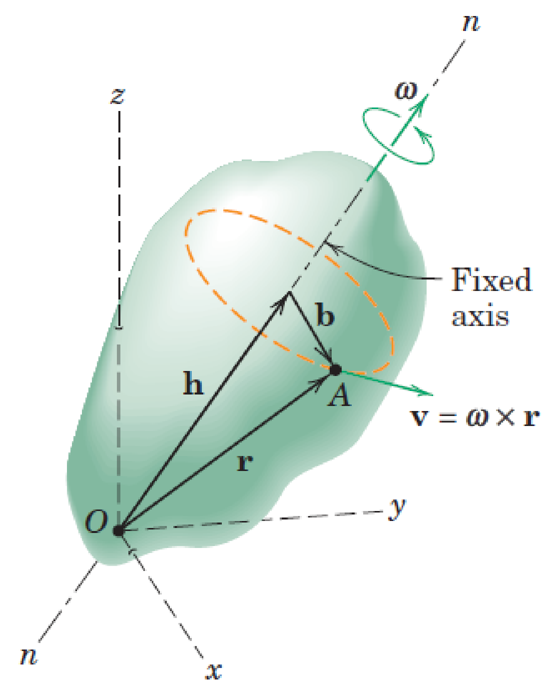
$$\boldsymbol{\alpha} = \boldsymbol{\Omega} \times \boldsymbol{\omega}$$

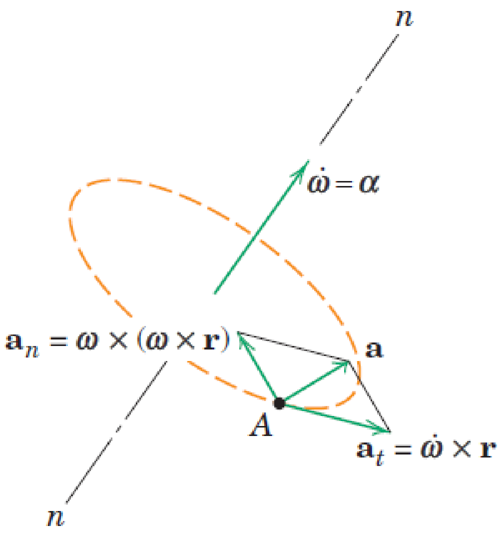
- The upper part of the figure relates the velocity of a point A on a rigid body to its position vector from O and the angular velocity of the body. The vectors $\boldsymbol{\alpha}$, $\boldsymbol{\omega}$ and $\boldsymbol{\Omega}$ in the lower figure bear exactly the same relationship to each other as do the vectors $\mathbf{v}$, $\mathbf{r}$ and $\boldsymbol{\omega}$ in the upper figure.



Angular Acceleration

Figure represent a rigid body rotating about a fixed point *O* with the instantaneous axis of rotation *n*-*n*, we see that the velocity $\mathbf{v}$ and acceleration $\boldsymbol{a}$ of any point A in the body are given by the same expressions as apply to the case in which the axis is fixed,





$$v = \omega \times r$$

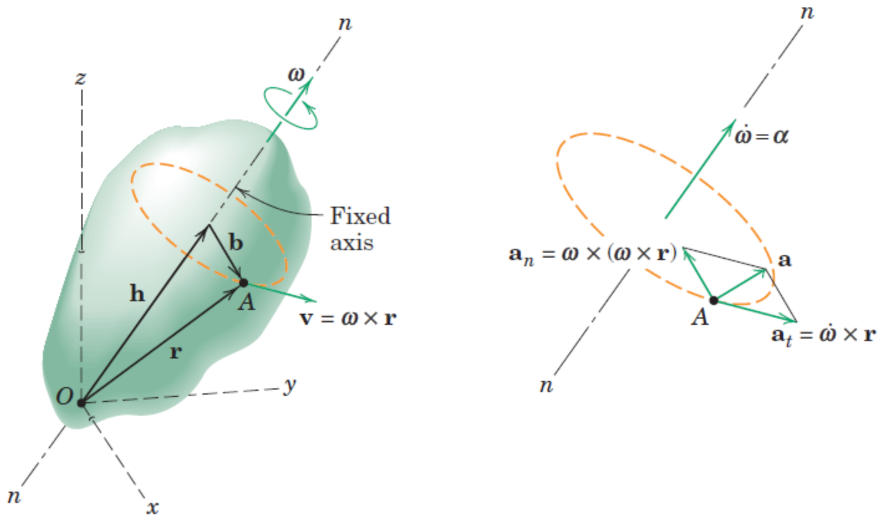
$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$

Angular Acceleration

- The one difference between the case of rotation about a fixed axis and rotation about a fixed point lies in the fact that for rotation about a fixed point, the angular acceleration $\boldsymbol{\alpha}$ will have a component normal to $\boldsymbol{\omega}$ due to the change in direction of $\boldsymbol{\omega}$ as well as a component in the direction of $\boldsymbol{\omega}$ to reflect any change in the magnitude of $\boldsymbol{\omega}$

- Although any point on the rotation axis n-n momentarily will have zero velocity, it will not have zero acceleration as long as it is changing its direction. On the other hand, for rotation about a fixed axis, $\boldsymbol{\alpha} = \dot{\boldsymbol{\omega}}$ has only the one component along the fixed axis to reflect the change in the magnitude of $\boldsymbol{\omega}$

- Furthermore, points which lie on the fixed rotation axis clearly have no velocity or acceleration.



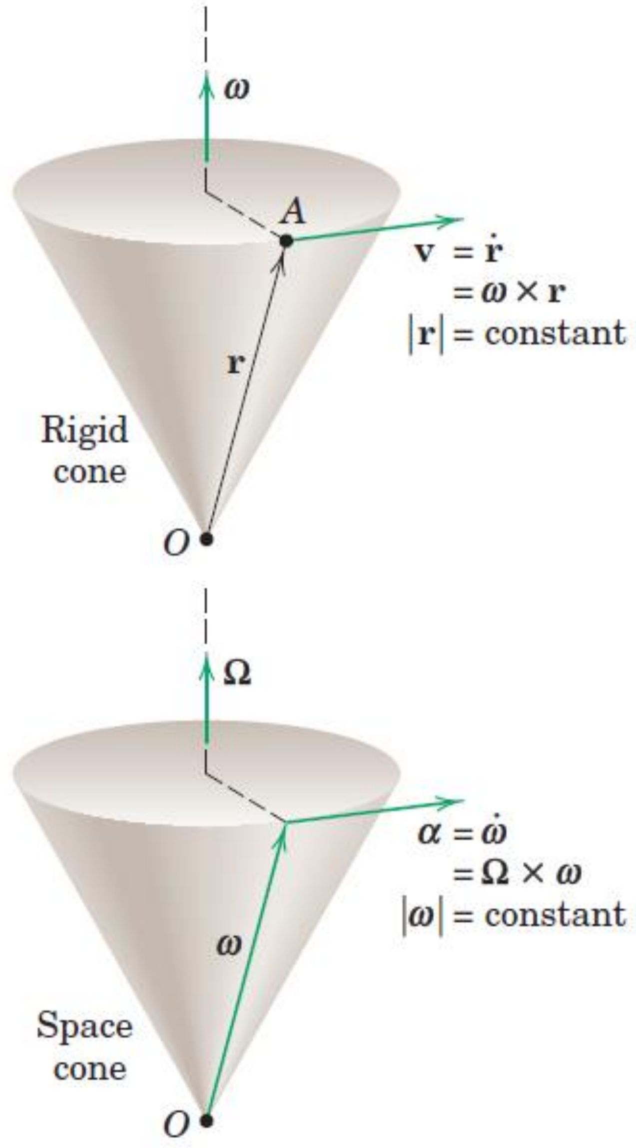
$$\mathrm{v} = \omega \times \mathrm{r}$$

$$a = \dot{\omega} \times r + \omega \times (\omega \times r)$$

Angular Acceleration

- Rotation is a function solely of angular change, so that the expressions for $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ do not depend on the fixity of the point around which rotation occurs

- Rotation may take place independently of the linear motion of the rotation point

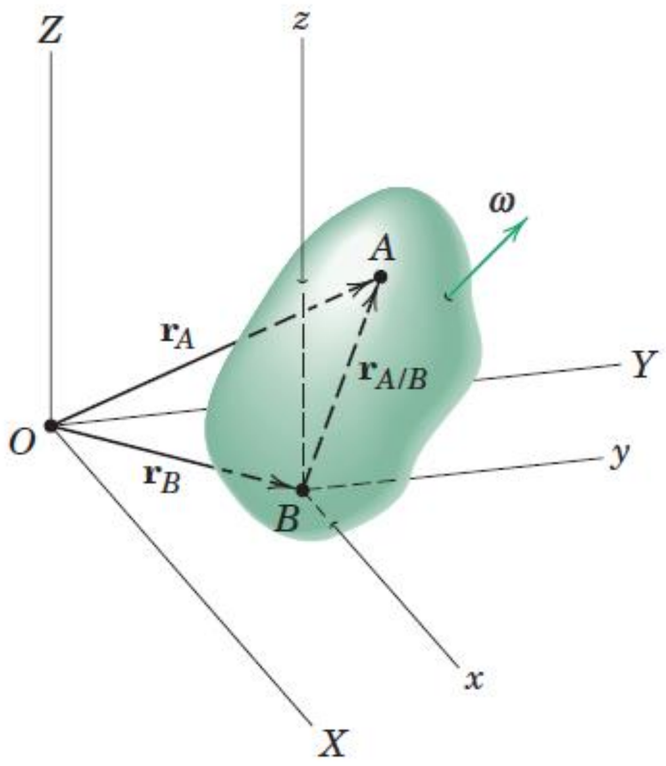


General Motion: Translating reference axes

- From an observer's position on *x-y-z*, the body appears to rotate about the point *B* and point *A* appears to lie on a spherical surface with *B* as the center.
- General motion as a translation of the body with the motion of *B* plus a rotation of the body about *B*.

$$\mathrm{v}_A = \mathrm{v}_B + \mathrm{v}_{A/B}$$

$$a_A = a_B + a_{A/B}$$

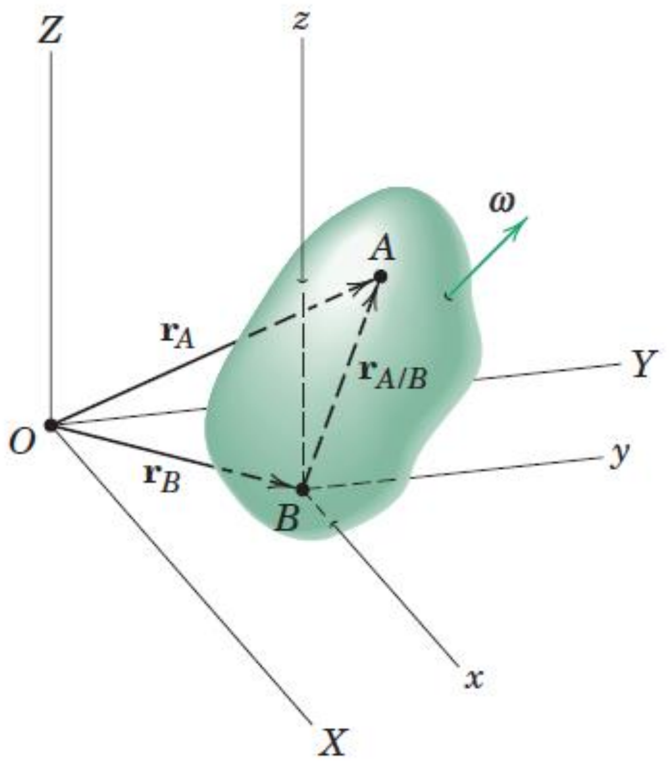


General Motion: Translating reference axes

- The relative-motion terms represent the effect of the rotation about B and are identical to the velocity and acceleration expressions discussed in the previous article for rotation of a rigid body about a fixed point.

$$\mathrm{v_A} = \mathrm{v_B} + \omega \times \mathrm{r_{A/B}}$$

$$a_A = a_B + \dot{\omega} \times r_{A/B} + \omega \times \left(\omega \times r_{A/B}\right)$$



- Point *A* is chosen as the reference point, the relative-motion equations

$$\mathrm{v_B} = \mathrm{v_A} + \omega \times \mathrm{r_{B/A}}$$

$$a_B = a_A + \dot{\omega} \times r_{B/A} + \omega \times \left(\omega \times r_{B/A}\right)$$

General Motion: Rotating reference axes

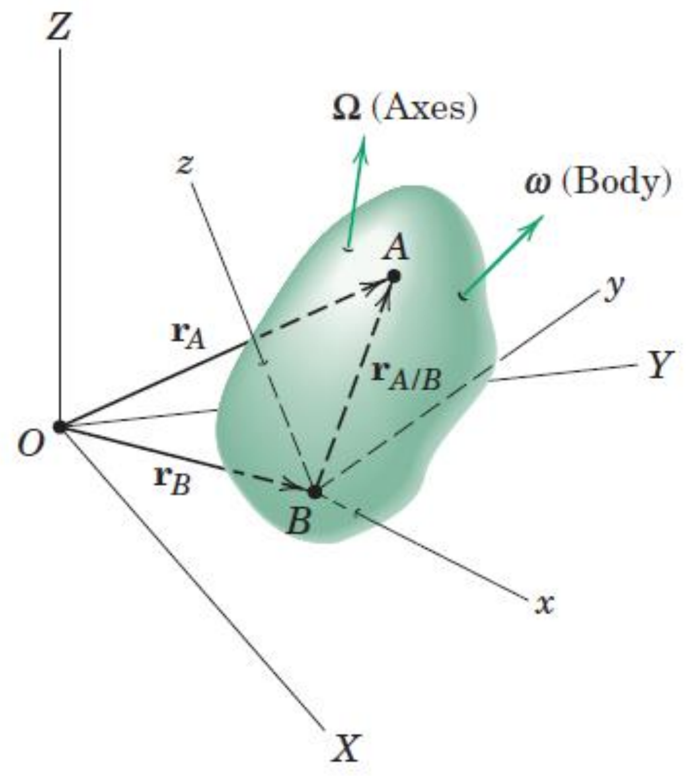
- Use of reference axes which rotate as well as translate

- To show reference axes whose origin is attached to the reference point B as before, but which rotate with an absolute angular velocity Ω which may be different from the absolute angular velocity ω of the body

$$\dot{\mathrm{i}} = \Omega \times \mathrm{i}$$

$$\dot{\mathrm{j}} = \Omega \times \mathrm{j}$$

$$\dot{\mathrm{k}} = \Omega \times \mathrm{k}$$



Time derivatives of the rotating unit vectors attached to *x*-*y*-*z*

General Motion: Rotating reference axes

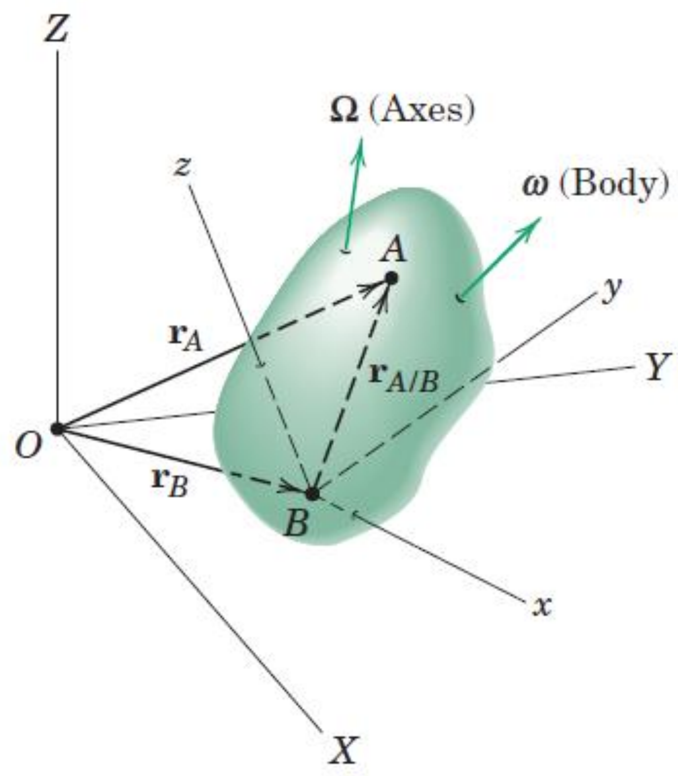
- Expression for the velocity and acceleration of point A

$$\mathbf{v}_A = \mathbf{v}_B + \Omega \times \mathbf{r}_{A/B} + \mathbf{v}_{rel}$$

$$\mathbf{a}_A = \mathbf{a}_B + \dot{\Omega} \times \mathbf{r}_{A/B} + \Omega \times \left(\Omega \times \mathbf{r}_{A/B}\right) + 2\Omega \times \mathbf{v}_{rel} + \mathbf{a}_{rel}$$

$$\mathbf{v}_{rel} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k}$$

$$\mathbf{a}_{rel} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k}$$



Velocity and acceleration of point *A* measured relative to *x-y-z* by an observer attached to *x-y-z*

General Motion: Rotating reference axes

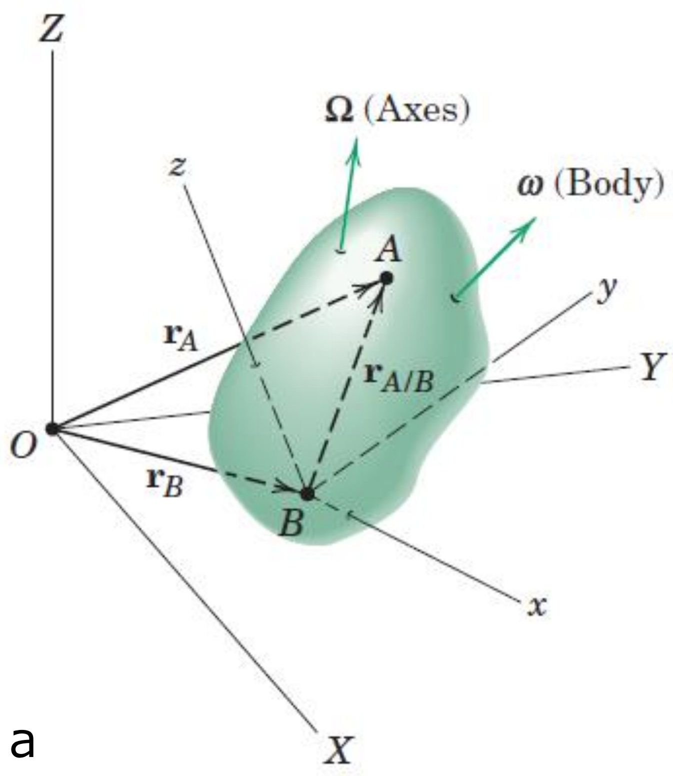
- Expression for the velocity and acceleration of point A

$$\mathrm{v}_A = \mathrm{v}_B + \Omega \times \mathrm{r}_{A/B} + \mathrm{v}_{rel}$$

$$\mathrm{a}_A = \mathrm{a}_B + \dot{\Omega} \times \mathrm{r}_{A/B} + \Omega \times (\Omega \times \mathrm{r}_{A/B}) + 2\Omega \times \mathrm{v}_{rel} + \mathrm{a}_{rel}$$

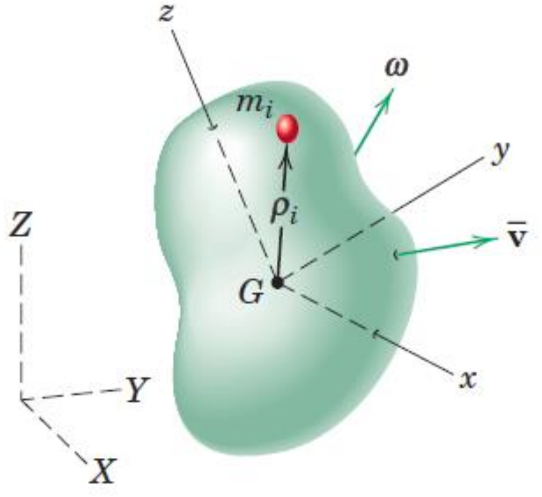
- Ω angular velocity of the axes and it may be different from the angular velocity ω of the body

- $\mathrm{r}_{A/B}$ remains constant in magnitude for points *A* and *B* fixed to a rigid body, but it will change direction with respect to *x-y-z* when the angular velocity of the axes is different from the angular velocity of the body



Angular Momentum

- Consider a rigid body moving with any general motion in space
- Axes x-y-z are attached to the body with origin at the mass center G
- The angular velocity of the body becomes the angular velocity of the x-y-z axes as observed from the fixed reference axes X-Y-Z.



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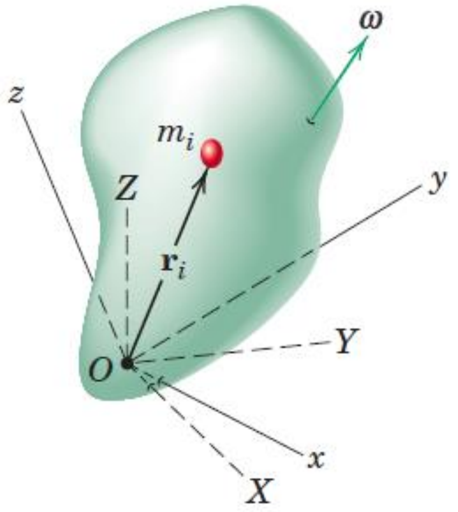


Angular Momentum

➢ Sum of the moments about G of the linear momenta of all elements of the body

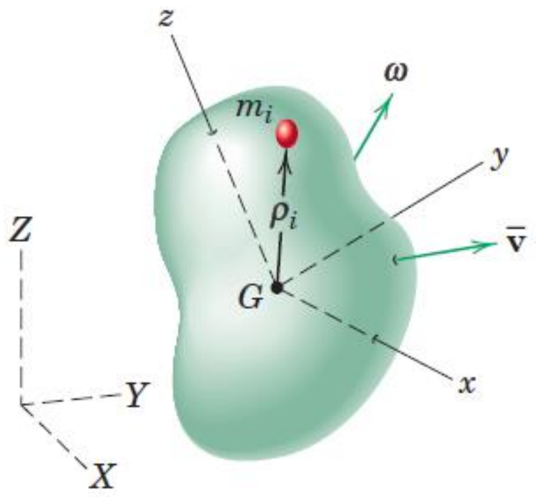
$$H_G = \sum(\rho_i \times m_i v_i)$$

$\boldsymbol{v_i}$ is the absolute velocity of mass element

$$v_i = \bar{v} + \omega \times \rho_i$$

$$H_G = \sum(\rho_i \times m_i(\bar{v} + \omega \times \rho_i))$$

$$H_G = -\bar{v} \times \sum m_i \rho_i + \sum \rho_i \times m_i(\omega \times \rho_i)$$



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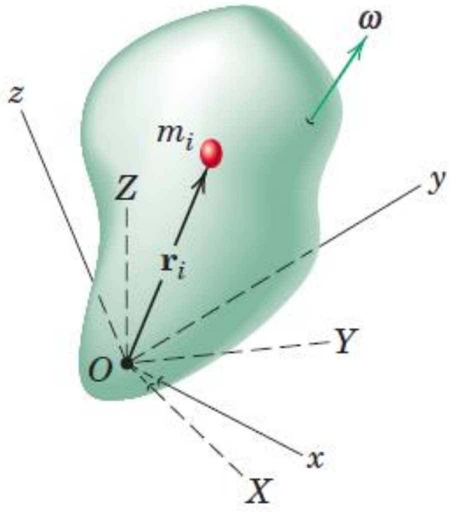


Angular Momentum

$$H_G = \sum(\rho_i \times m_i v_i)$$

$$H_G = -\bar{v} \times \sum m_i \rho_i + \sum \rho_i \times m_i(\omega \times \rho_i)$$

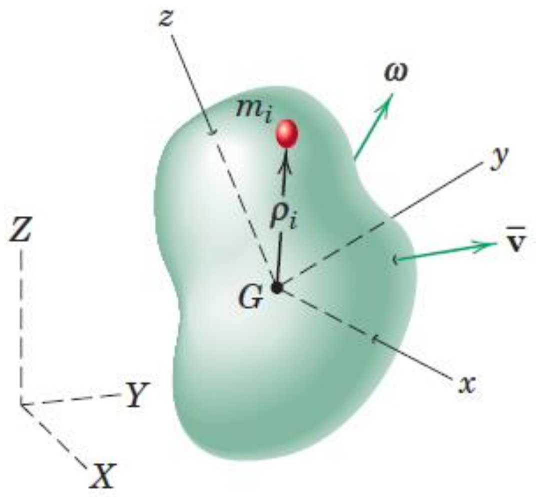
With the origin at the mass center G,

$$\sum m_i \rho_i = m\bar{\rho} = 0$$

dm for m_i and ρ for ρ_i

$$H_G = \int [\rho \times (\omega \times \rho)]\, \mathrm{d}m$$

$$H_0 = \int [r \times (\omega \times r)]\, \mathrm{d}m$$



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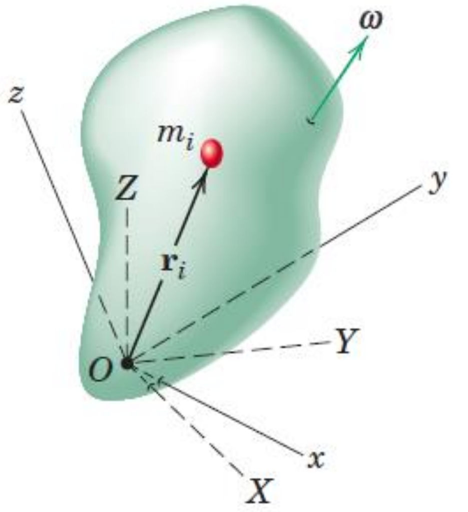


Moments and Product of Inertia

$$\mathbf{H_G} = \int [\rho \times (\omega \times \rho)]\, \mathrm{d}m$$

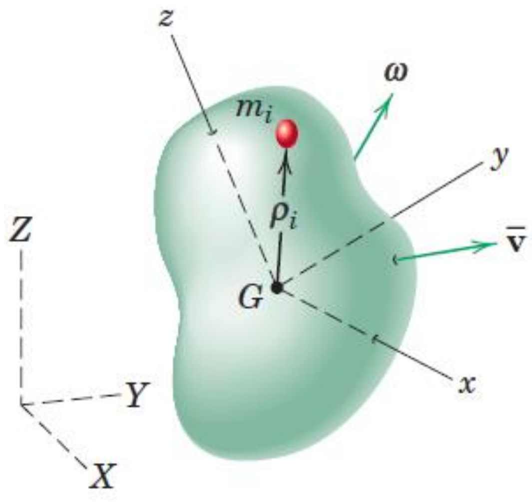
$$\mathbf{H_0} = \int [r \times (\omega \times r)]\, \mathrm{d}m = \int d\mathrm{H}$$

$$r = xi + yj + zk \qquad \omega = \omega_x i + \omega_y j + \omega_z k$$

$$d\mathbf{H} = \mathrm{i}\left[(y^2 + z^2)\omega_x - xy\omega_y - xz\omega_z\right] \mathrm{d}m$$

$$+j\left[-yx\omega_x + (z^2 + x^2)\omega_y - yz\omega_z\right] \mathrm{d}m$$

$$+k\left[-zx\omega_x - zy\omega_y + (x^2 + y^2)\omega_z\right] \mathrm{d}m$$



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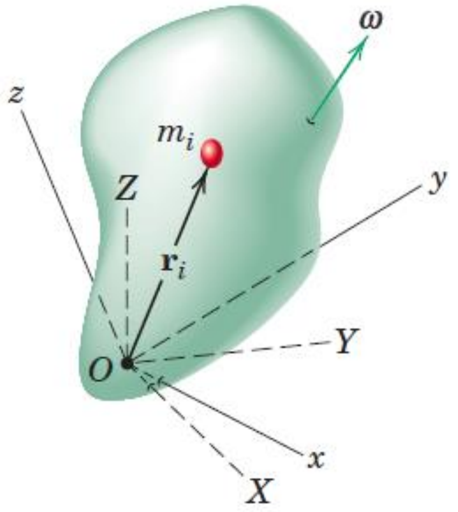
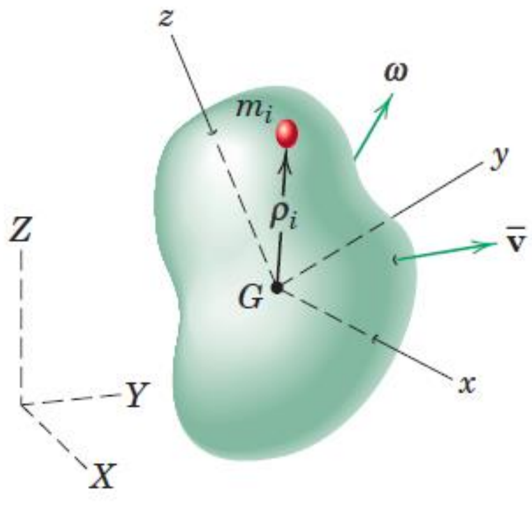


Moments and Product of Inertia

$$d\mathbf{H} = \mathrm{i}\left[(y^2+z^2)\omega_x - xy\omega_y - xz\omega_z\right]\mathrm{dm}$$
$$+j\left[-yx\omega_x + (z^2+x^2)\omega_y - yz\omega_z\right]\mathrm{dm}$$
$$+k\left[-zx\omega_x - zy\omega_y + (x^2+y^2)\omega_z\right]\mathrm{dm}$$

$$I_{xx} = \int (y^2+z^2)\,\mathrm{dm} \qquad I_{xy} = \int xy\,\mathrm{dm}$$
$$I_{yy} = \int (z^2+x^2)\,\mathrm{dm} \qquad I_{xz} = \int xz\,\mathrm{dm}$$
$$I_{zz} = \int (x^2+y^2)\,\mathrm{dm} \qquad I_{yz} = \int yz\,\mathrm{dm}$$



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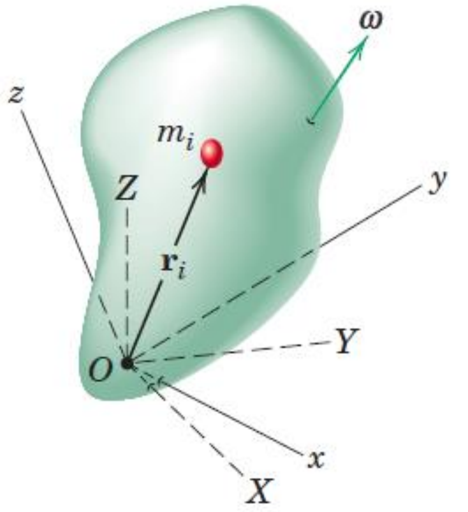
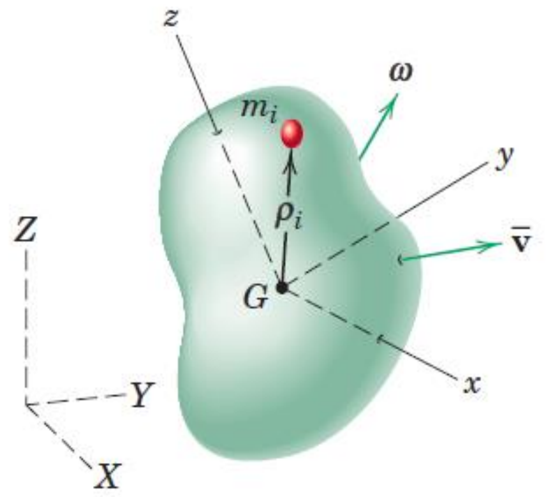


Moments and Product of Inertia

$$I_{xx} = \int (y^2 + z^2)\,\mathrm{d}m \qquad I_{xy} = \int xy\,\mathrm{d}m$$

$$I_{yy} = \int (z^2 + x^2)\,\mathrm{d}m \qquad I_{xz} = \int xz\,\mathrm{d}m$$

$$I_{zz} = \int (x^2 + y^2)\,\mathrm{d}m \qquad I_{yz} = \int yz\,\mathrm{d}m$$



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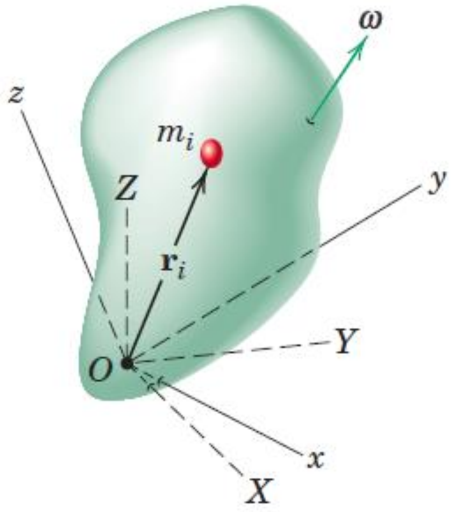


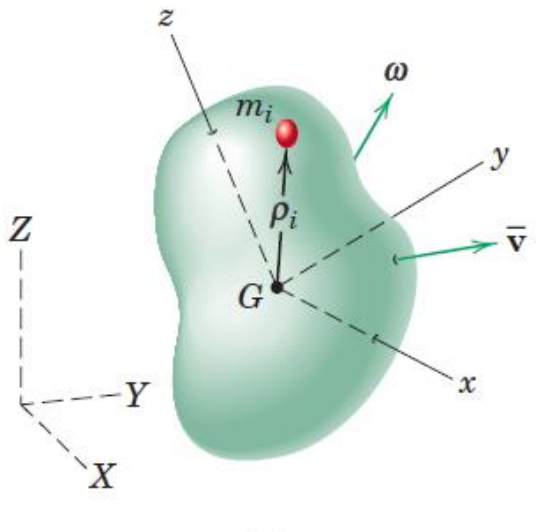
Moments of inertia of the body about the respective axes,
$\boldsymbol{I_{xx}, I_{yy}, I_{zz}}$
Products of inertia with respect to the coordinate axes
$\boldsymbol{I_{xy}, I_{xz}, I_{yz}}$

These quantities describe the manner in which the mass of a rigid body is distributed with respect to the chosen axes.

Moments and Product of Inertia

$$\mathbf{H} = \mathrm{i}\left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$
$$+j\left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$
$$+k\left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

$$I_{xx} = \int (y^2 + z^2)\,\mathrm{d}m \qquad I_{xy} = \int xy\,\mathrm{d}m$$
$$I_{yy} = \int (z^2 + x^2)\,\mathrm{d}m \qquad I_{xz} = \int xz\,\mathrm{d}m$$
$$I_{zz} = \int (x^2 + y^2)\,\mathrm{d}m \qquad I_{yz} = \int yz\,\mathrm{d}m$$



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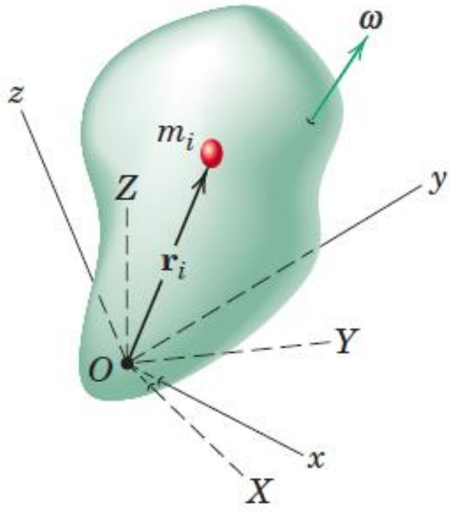


Moments and Product of Inertia

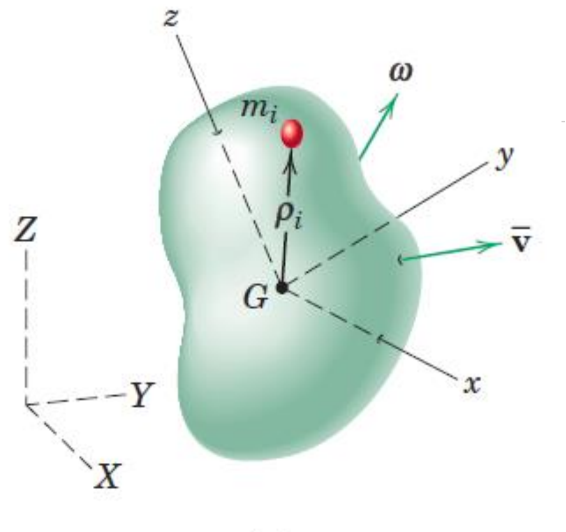
$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + (I_{yy})\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

- General expression for the angular momentum either about the mass center *G* or about a fixed point *O* for a rigid body rotating with an instantaneous angular velocity, ω

- In each of the two cases represented, the reference axes x-y-z are attached to the rigid body. This attachment makes the moment-of-inertia integrals and the product-of-inertia integrals invariant with time.



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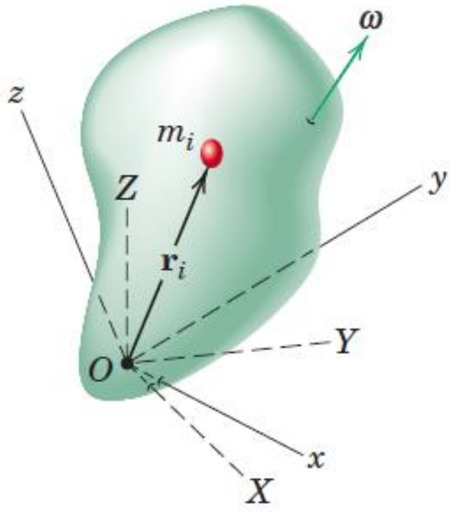


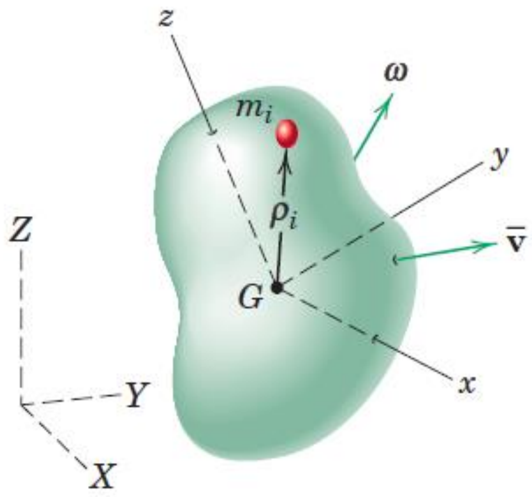
Moments and Product of Inertia

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + (I_{yy})\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

- If the x-y-z axes were to rotate with respect to an irregular body, then these inertia integrals would be functions of the time, which would introduce an undesirable complexity into the angular momentum relations.



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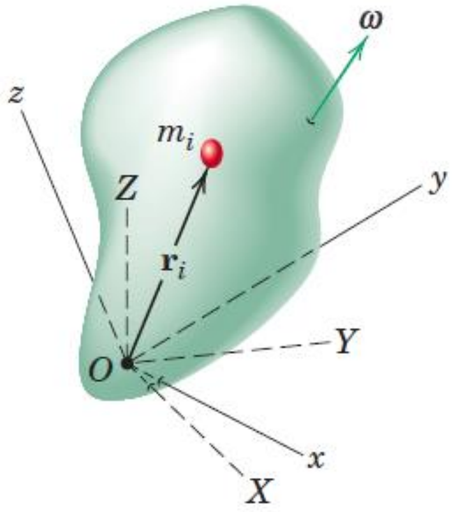
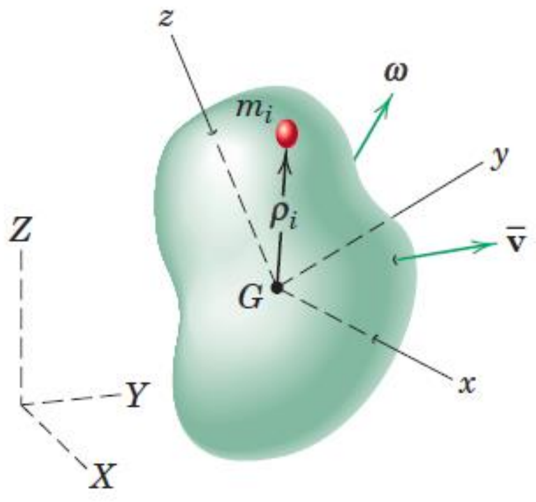


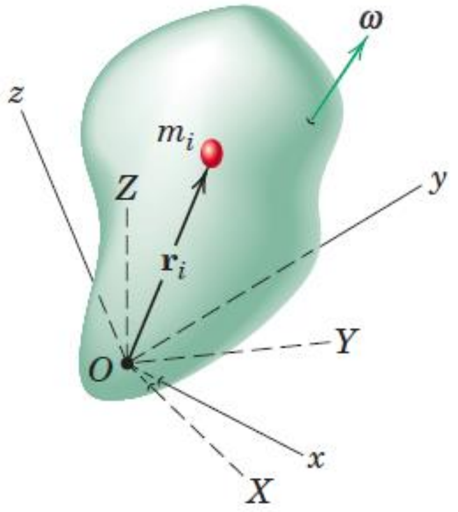
Inertia Matrix or Inertia tensor

$$\begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix}$$



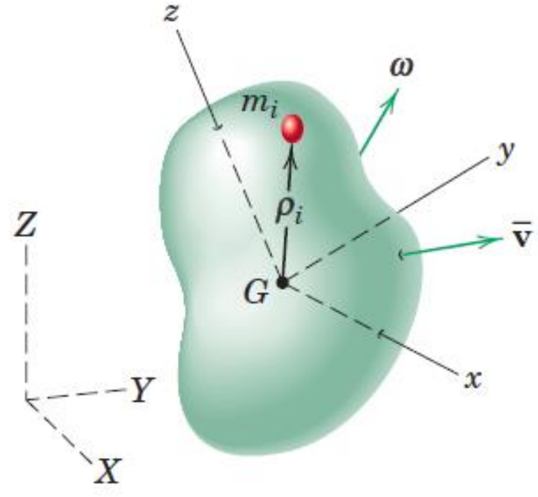
(a)

- As we change the orientation of the axes relative to the body, the moments and products of inertia will also change in value. There is one unique orientation of axes x-y-z for a given origin for which the products of inertia vanish and the moments of inertia I_{xx}, I_{yy}, I_{zz} take on stationary values.



Principal moments of inertia

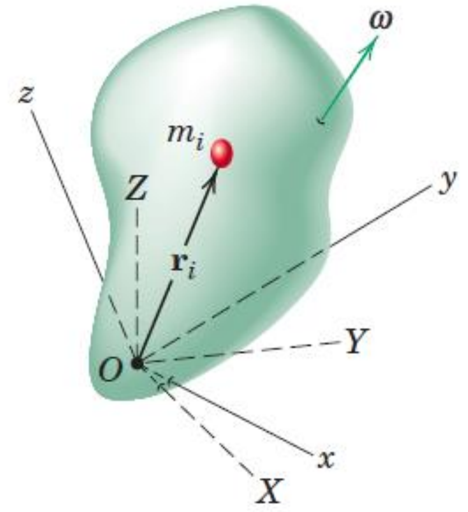
$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$



(a)

- The axes x-y-z for which the products of inertia vanish are called the **principal axes of inertia**

- I_{xx}, I_{yy}, and I_{zz} are called the **principal moments of inertia**



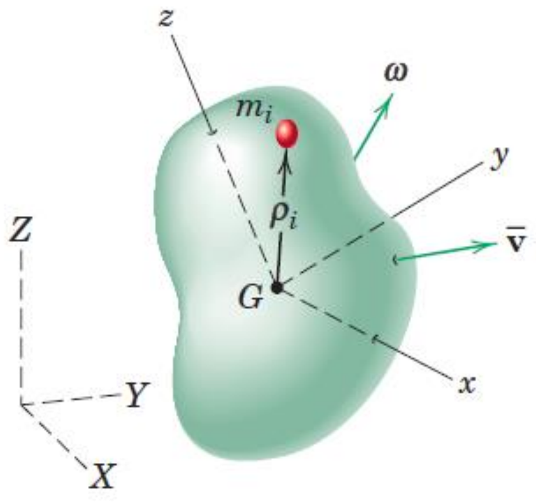
- The principal moments of inertia for a given origin represent the maximum, the minimum, and an intermediate value of the moments of inertia

Angular Momentum

$$\begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}$$

- The axes x-y-z for which the products of inertia vanish are called the **principal axes of inertia**
- I_{xx}, I_{yy}, and I_{zz} are called the **principal moments of inertia**

$$\mathbf{H} = I_{xx}\omega_x\mathbf{i} + I_{yy}\omega_y\mathbf{j} + I_{zz}\omega_z\mathbf{k}$$



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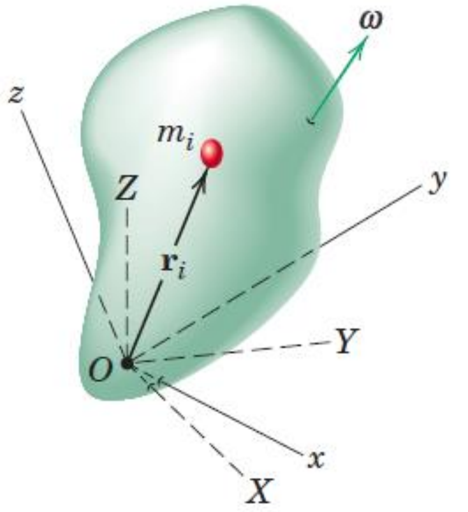
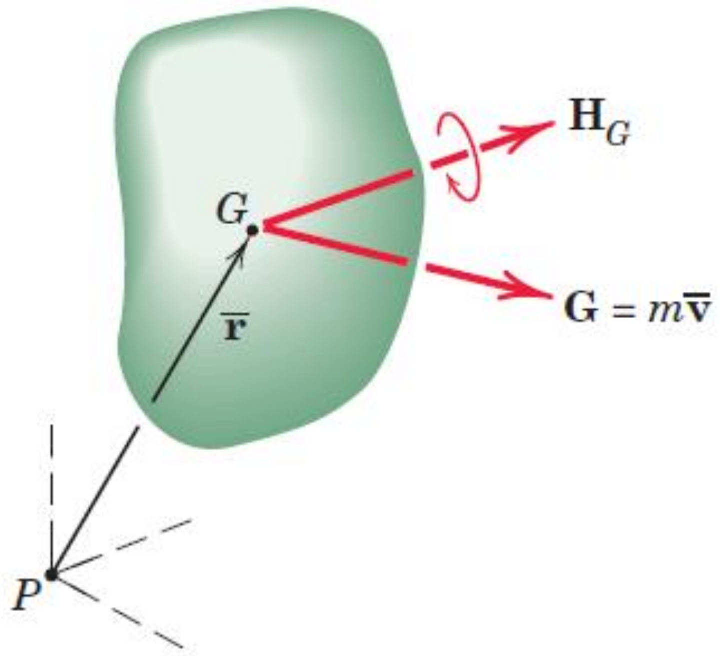


Transfer Principle for Angular Momentum

- The momentum properties of a rigid body may be represented by the resultant linear-momentum vector $\boldsymbol{G} = m\bar{\mathbf{v}}$ through the mass center and the resultant angular-momentum vector $\mathbf{H_G}$ about the mass center. Although $\mathbf{H_G}$ has the properties of a free vector, we represent it through G for convenience
- These vectors have properties analogous to those of a force and a couple. Thus, the angular momentum about any point P equals the free vector $\mathbf{H_G}$ plus the moment of the linear-momentum vector $\boldsymbol{G}$ about P
- Transfer theorem for angular momentum

$$H_P = H_G + \bar{r} \times G$$



Kinetic Energy

$$T = \frac{1}{2} m\bar{v}^2 + \sum \frac{1}{2} m_i |\dot{\rho}_i|^2$$

- $\bar{\mathbf{v}}$ is velocity of mass center G and ρ_i is position vector of mass element m_i with respect to mass center

- First term as the kinetic energy due to the translation of the system and the second term as the kinetic energy associated with the motion relative to the mass center

$$\frac{1}{2} m\bar{v}^2 = \frac{1}{2} m\dot{\bar{\boldsymbol{r}}}.\dot{\bar{\boldsymbol{r}}} = \frac{1}{2}\bar{\mathbf{v}}.\mathbf{G}$$

- $\mathbf{G}$ is linear momentum of the body

Kinetic Energy

$$T = \frac{1}{2}m\bar{v}^2 + \sum \frac{1}{2}m_i|\dot{\rho}_i|^2$$

$$\sum \frac{1}{2}m_i|\dot{\rho}_i|^2 = \sum \frac{1}{2}m_i(\omega \times \rho_i).(\omega \times \rho_i)$$

$$\mathbf{P} \times \mathbf{Q}.\mathbf{R} = \mathbf{P}.\mathbf{Q} \times \mathbf{R}$$

$$\sum \frac{1}{2}m_i(\omega \times \rho_i).(\omega \times \rho_i) = \sum \frac{1}{2}m_i\omega.\rho_i \times (\omega \times \rho_i) = \omega.\sum \frac{1}{2}m_i\rho_i \times (\omega \times \rho_i)$$

$$\omega.\sum \frac{1}{2}m_i\rho_i \times (\omega \times \rho_i) = \frac{1}{2}\omega.\int \rho \times (\omega \times \rho)dm = \frac{1}{2}\omega.\int \rho \times (\omega \times \rho)dm = \frac{1}{2}\boldsymbol{\omega}.\mathbf{H_G}$$

Kinetic Energy

General expression for the kinetic energy of a rigid body moving with mass-center velocity $\bar{\mathbf{v}}$ and angular velocity $\boldsymbol{\omega}$ is

$$T = \frac{1}{2}m\bar{v}^2 + \sum \frac{1}{2}m_i|\dot{\rho}_i|^2 = \frac{1}{2}\bar{\mathbf{v}}.\mathbf{G} + \frac{1}{2}\boldsymbol{\omega}.\mathbf{H_G}$$

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

Kinetic Energy

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$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\left(\bar{I}_{xx}\omega_x^2 + \bar{I}_{yy}\omega_y^2 + \bar{I}_{zz}\omega_z^2\right) - \left(\bar{I}_{xy}\omega_x\omega_y + \bar{I}_{xz}\omega_x\omega_z + \bar{I}_{yz}\omega_y\omega_z\right)$$

If the axes coincide with principal axes of inertia,

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2}\left(\bar{I}_{xx}\omega_x^2 + \bar{I}_{yy}\omega_y^2 + \bar{I}_{zz}\omega_z^2\right)$$

Kinetic Energy

When a rigid body is pivoted about a fixed point O or when there is a point O in the body which momentarily has zero velocity, the kinetic energy, T

$$T = \sum \frac{1}{2} m_i \dot{\boldsymbol{r}}_i . \dot{\boldsymbol{r}}_i$$

This expression reduces to

$$T = \frac{1}{2} \boldsymbol{\omega} . \mathbf{H}_{\mathrm{o}}$$

where $\mathbf{H}_{\mathrm{o}}$ is the angular momentum about O.

Momentum Equations

General linear- and angular-momentum equations for a system of constant mass

$$\sum F = \dot{G} \qquad \sum M = \dot{H}$$

In the derivation of the moment principle, the derivative of H was taken with respect to an absolute coordinate system.

When H is expressed in terms of components measured relative to a moving coordinate system *x*-*y*-*z* which has an angular velocity Ω, the moment relation becomes

$$\sum M = \left(\frac{\mathrm{d}H}{\mathrm{d}t}\right)_{xyz} + \Omega \times \mathrm{H}$$

Momentum Equations

When H is expressed in terms of components measured relative to a moving coordinate system *x*-*y*-*z* which has an angular velocity Ω, the moment relation becomes

$$\sum \boldsymbol{M} = \left(\frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}t}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

The terms in parentheses represent that part of $\dot{\boldsymbol{H}}$ due to the change in magnitude of the components of **H**, and the cross-product term represents that part due to the changes in direction of the components of **H**.

Momentum Equations

$$\sum \boldsymbol{M} = \left(\frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}t}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H_y\Omega_z} + \mathrm{H_z\Omega_y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H_z\Omega_x} + \mathrm{H_x\Omega_z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H_x\Omega_y} + \mathrm{H_y\Omega_x}\right)\mathrm{k}$$

Most general form of the moment equation about a fixed point *O* or about the mass center *G*
Ω's are the angular velocity components of rotation of the reference axes
In the *H* components of a rigid body, ω the angular velocity of the body is used

Momentum Equations

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H}_\mathrm{y}\Omega_\mathrm{z} + \mathrm{H}_\mathrm{z}\Omega_\mathrm{y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H}_\mathrm{z}\Omega_\mathrm{x} + \mathrm{H}_\mathrm{x}\Omega_\mathrm{z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H}_\mathrm{x}\Omega_\mathrm{y} + \mathrm{H}_\mathrm{y}\Omega_\mathrm{x}\right)\mathrm{k}$$

Most general form of the moment equation about a fixed point *O* or about the mass center *G*.
Ω's are the angular velocity components of rotation of the reference axes
In the *H* components of a rigid body, ω the angular velocity of the body is used

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

Momentum Equations

$$\sum \boldsymbol{M} = (\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}) + \mathbf{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = (\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H}_\mathrm{y}\Omega_\mathrm{z} + \mathrm{H}_\mathrm{z}\Omega_\mathrm{y})\mathrm{i} + (\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H}_\mathrm{z}\Omega_\mathrm{x} + \mathrm{H}_\mathrm{x}\Omega_\mathrm{z})\mathrm{j} + (\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H}_\mathrm{x}\Omega_\mathrm{y} + \mathrm{H}_\mathrm{y}\Omega_\mathrm{x})\mathrm{k}$$

- Axes x-y-z attached to rigid body
- Under these conditions, when expressed in the x-y-z coordinates, the moments and products of inertia are invariant with time and $\Omega = \omega$

$$\sum M_x = \dot{H}_x - H_y\omega_z + H_z\omega_y$$

$$\sum M_y = \dot{H}_y - H_z\omega_x + H_x\omega_z \qquad \sum M_z = \dot{H}_z - H_x\omega_y + H_y\omega_x$$

- General moment equations for rigid-body motion with axes *attached to the body*
- They hold with respect to axes through a fixed point *O* or through the mass center *G*

Euler's Equations

$$\sum \mathrm{M} = (\dot{\mathrm{H}}_x - \mathrm{H}_y\Omega_z + \mathrm{H}_z\Omega_y)\mathrm{i} + (\dot{\mathrm{H}}_y - \mathrm{H}_z\Omega_x + \mathrm{H}_x\Omega_z)\mathrm{j} + (\dot{\mathrm{H}}_z - \mathrm{H}_x\Omega_y + \mathrm{H}_y\Omega_x)\mathrm{k}$$

For any origin fixed to a rigid body, there are three principal axes of inertia with respect to which the products of inertia vanish.
If the reference axes coincide with the principal axes of inertia with origin at the mass center *G* or at a point *O* fixed to the body and fixed in space, the factors I_{xy}, I_{yz} *and* I_{xz} will be zero. Then,

$$\sum M_x = I_{xx}\dot{\omega}_x - (I_{yy} - I_{zz})\omega_y\omega_z$$

$$\sum M_y = I_{yy}\dot{\omega}_y - (I_{zz} - I_{xx})\omega_z\omega_x$$

$$\sum M_z = I_{zz}\dot{\omega}_z - (I_{xx} - I_{yy})\omega_x\omega_y$$

- Euler's Equations
- Useful in the study of rigid-body motion

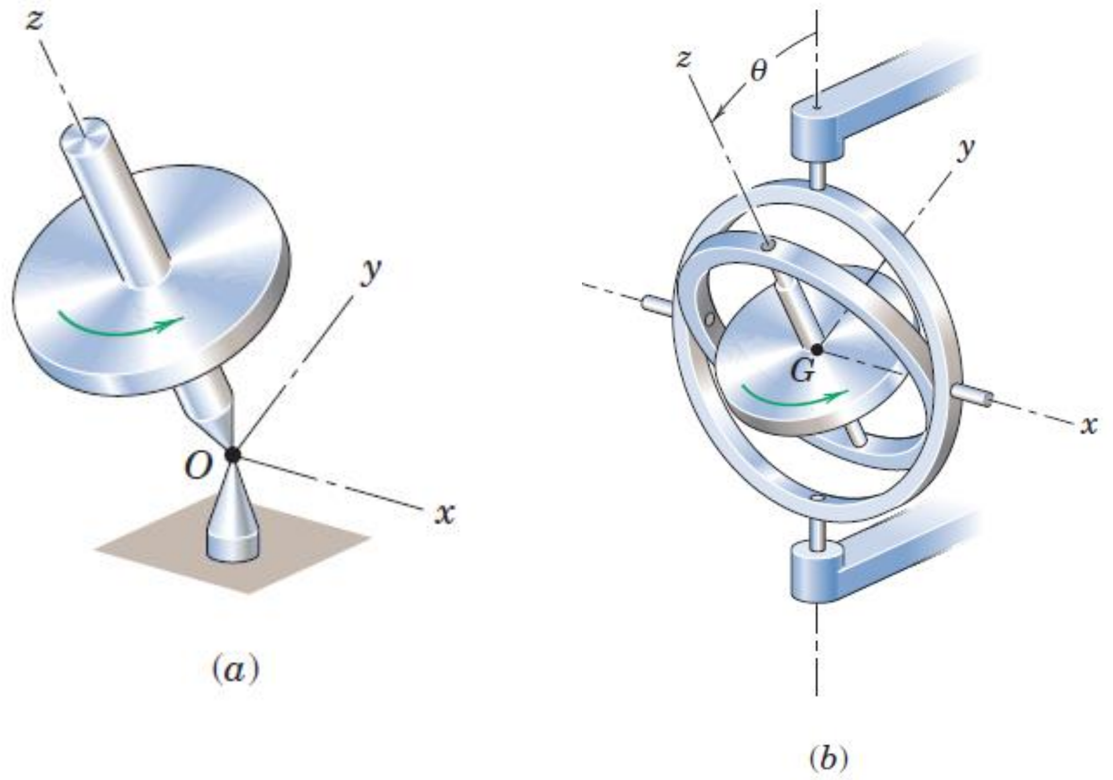
Gyroscopic Motion: Steady Precession

Gyroscopic motion: This motion occurs whenever the axis about which a body is spinning is itself rotating about another axis

Special Case: Gyroscopic motion when the **axis of a rotor spinning at constant speed turns** (precesses) about **another axis at a steady rate**

Gyroscopic Motion: Engineering Applications

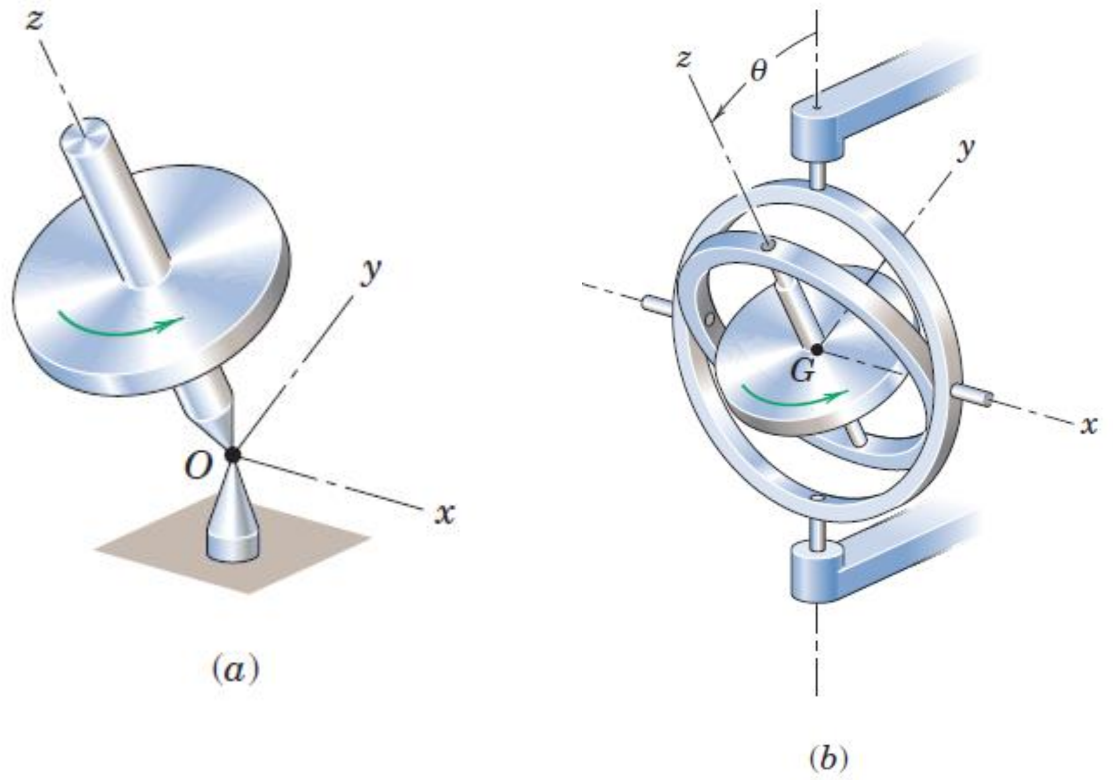
- The gyroscope has important engineering applications. With a mounting in gimbal rings, the gyro is free from external moments, and its axis will retain a fixed direction in space regardless of the rotation of the structure to which it is attached. In this way, the gyro is used for inertial guidance systems and other directional control devices.



Gyroscopic Motion: Engineering Applications

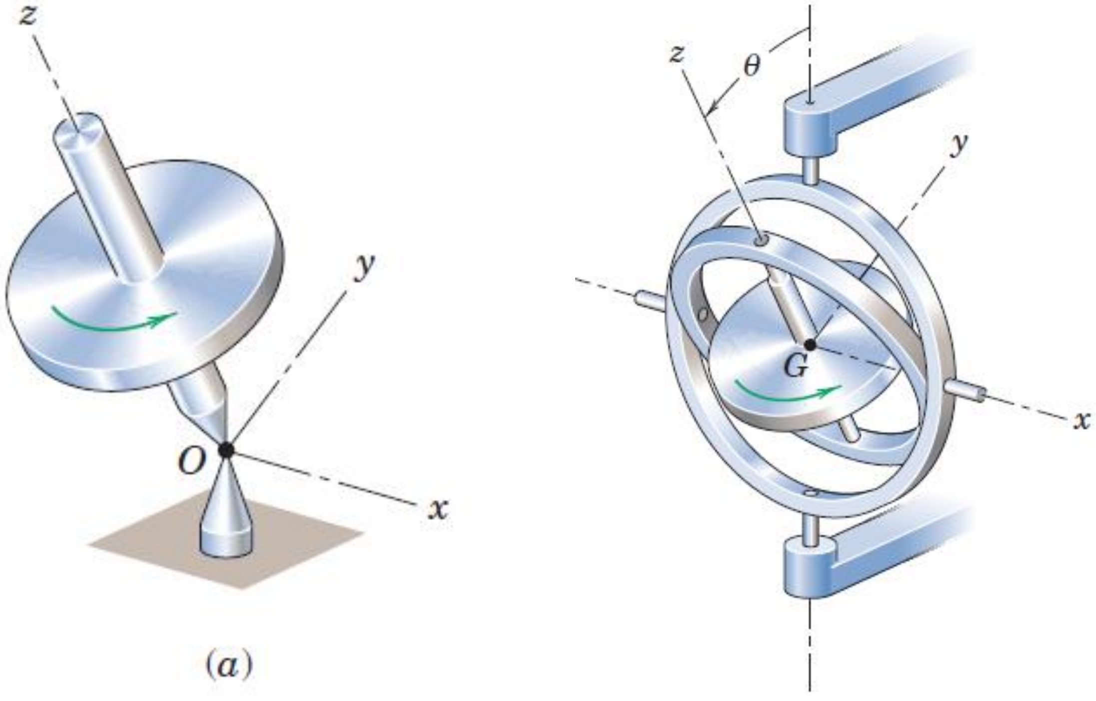
- With the addition of a pendulous mass to the inner gimbal ring, the earth's rotation causes the gyro to precess so that the spin axis will always point north, and this action forms the basis of the gyro compass

- The gyroscope has also found important use as a stabilizing device. The controlled precession of a large gyro mounted in a ship is used to produce a gyroscopic moment to counteract the rolling of a ship at sea



Gyroscopic Motion: Engineering Applications

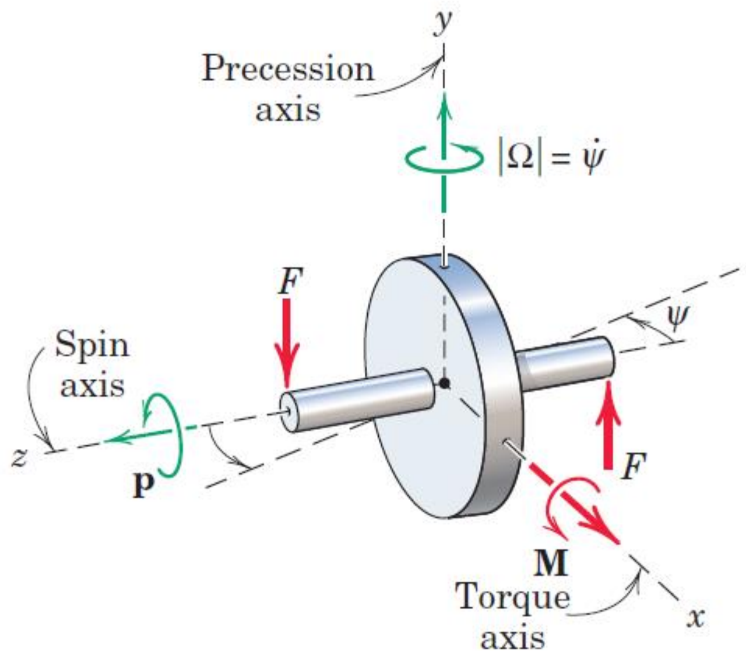
- The gyroscopic effect is also an extremely important consideration in the design of bearings for the shafts of rotors which are subjected to forced precessions



(a) (b)

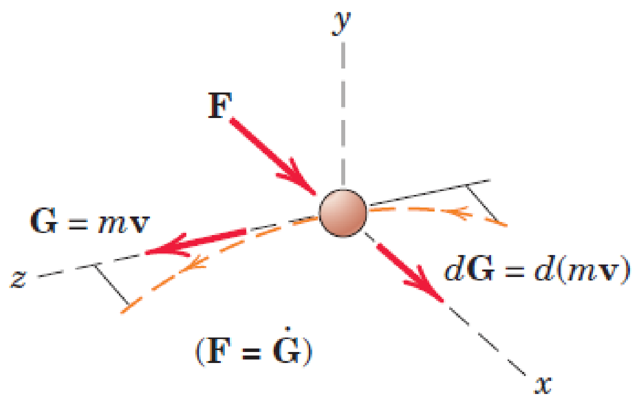
Gyroscopic Motion: Simplified Analysis

- Figure shows a symmetrical rotor spinning about the z-axis with a large angular velocity **p**, known as the **spin velocity**
- If we apply two forces **F** to the rotor axle to form a couple **M** whose vector is directed along the x-axis, we will find that the rotor shaft rotates in the x-z plane about the y-axis in the sense indicated, with a relatively slow angular velocity $\Omega = \dot{\psi}$ known as the **precession velocity**
- we identify the spin axis (p), the torque axis (M), and the precession axis (Ω), where the usual right-hand rule identifies the sense of the rotation vectors
- The rotor shaft does not turn about the x-axis in the sense of **M**, as it would if the rotor were not spinning



Gyroscopic Motion: Simplified Analysis

- To aid understanding of this phenomenon, a direct analogy may be made between the rotation vectors and the vectors which describe the curvilinear motion of a particle
- Figure shows a particle of mass *m* moving in the x-z plane with constant speed $|\mathbf{v}| = v$.



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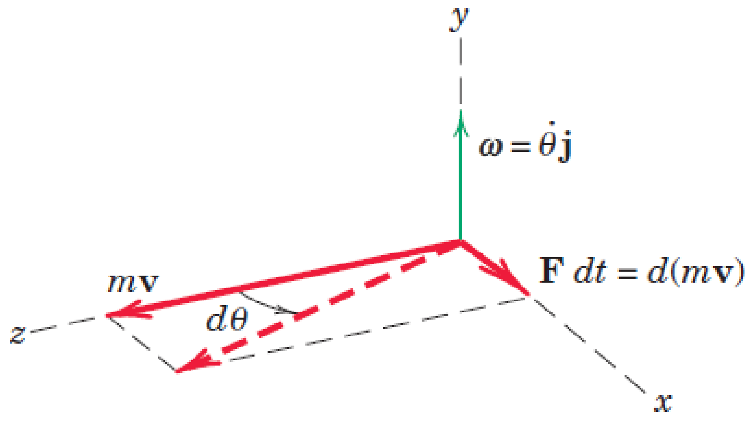

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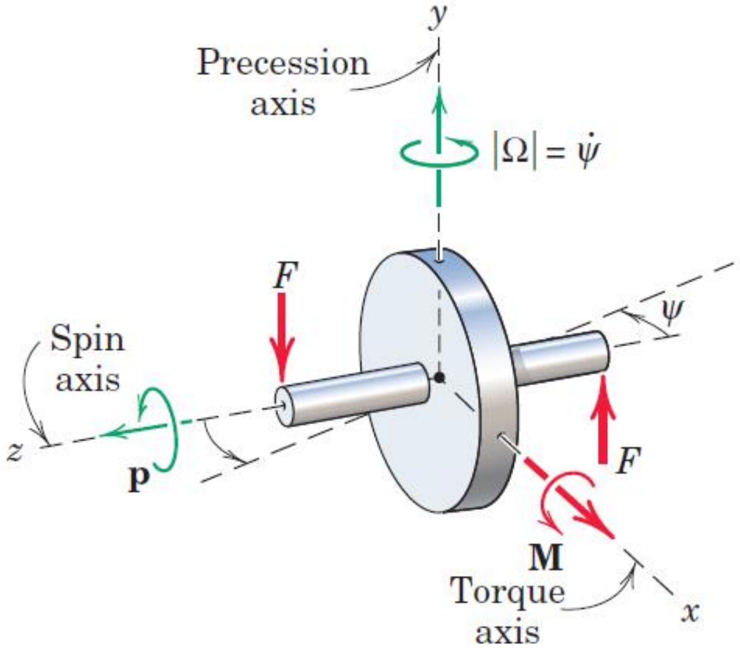
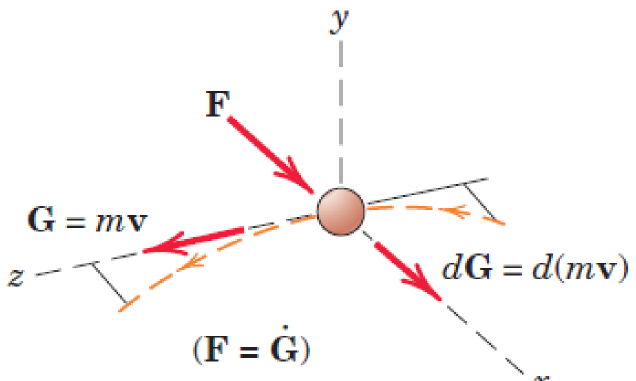
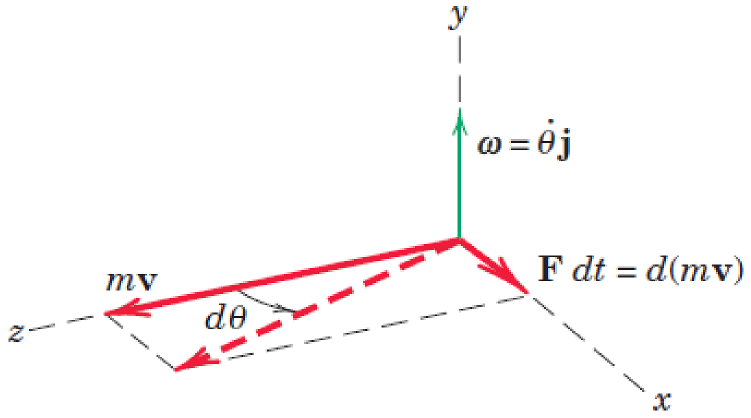


Gyroscopic Motion: Simplified Analysis

- The application of a force **F** normal to its linear momentum $\boldsymbol{G} = m\mathbf{v}$ causes a change $d\boldsymbol{G} = d(m\mathbf{v})$ in its momentum
- $d\boldsymbol{G}$, and thus $d\mathbf{v}$, is a vector in the direction of the normal force **F** according to Newton's second law $\boldsymbol{F} = \dot{\boldsymbol{G}}$ which may be written as $\boldsymbol{F}dt = d\boldsymbol{G}$
- In the limit, $\tan d\theta = d\theta = \mathbf{F}\,dt/m\mathbf{v}$ or $\mathbf{F} = m\mathbf{v}\dot{\boldsymbol{\theta}} = m\mathbf{v}\boldsymbol{\omega}$ In vector notation with $\boldsymbol{\omega} = \dot{\theta}\mathbf{j}$, the force becomes $\mathbf{F} = m\boldsymbol{\omega} \times \mathbf{v}$



(a)



(b)

Gyroscopic Motion: Simplified Analysis

- The application of a force **F** normal to its linear momentum $\boldsymbol{G} = m\mathbf{v}$ causes a change $d\boldsymbol{G} = d(m\mathbf{v})$ in its momentum
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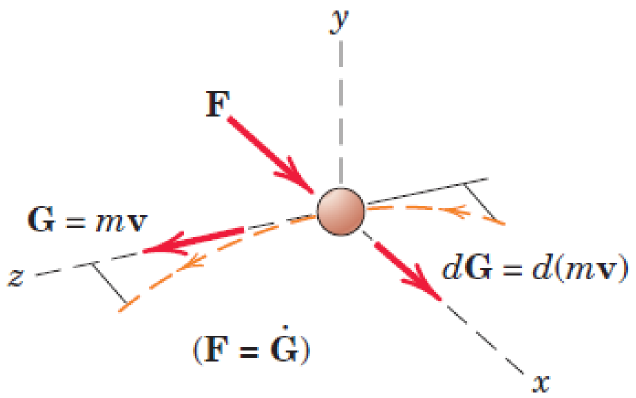
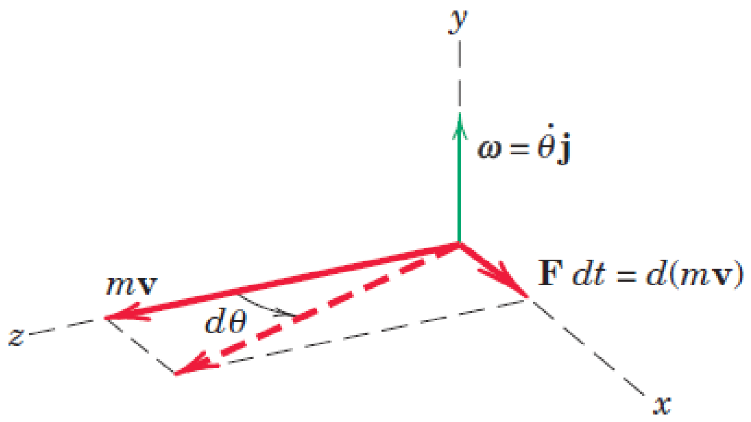


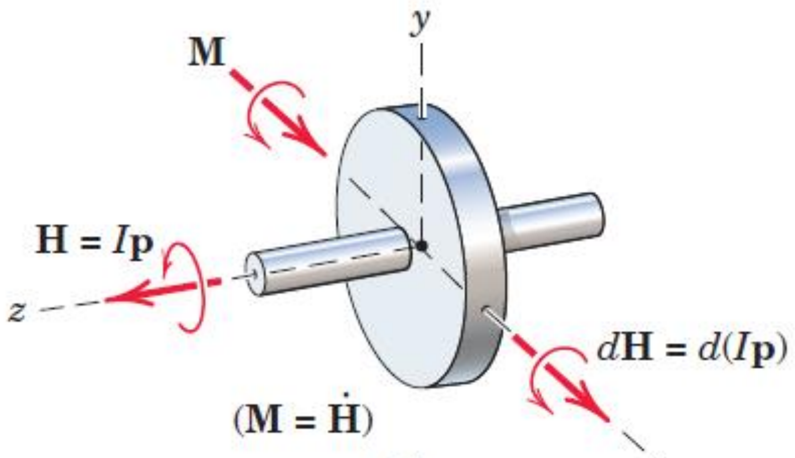
(a)



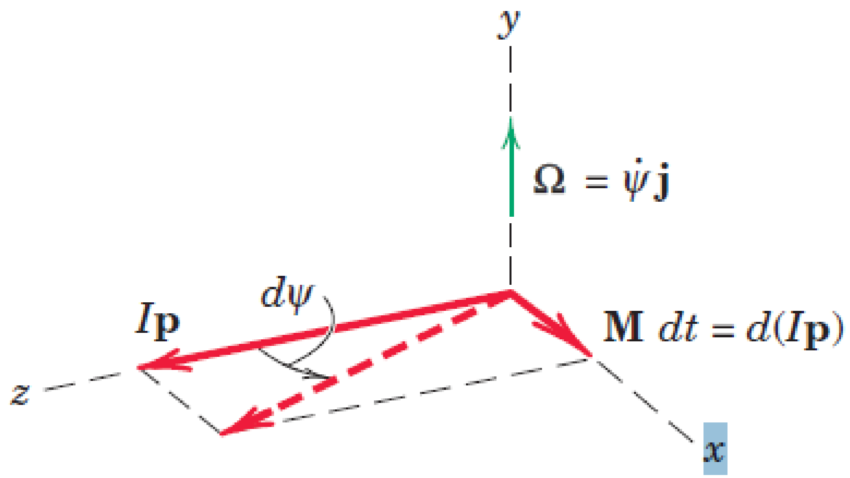
(b)

Gyroscopic Motion: Simplified Analysis

- Recall now the analogous equation $\mathbf{M} = \dot{\mathbf{H}}$ for any prescribed mass system, rigid or nonrigid, referred to its mass center or to a fixed point O
- Apply this relation to our symmetrical rotor, as shown in figure. For a high rate of spin $\mathbf{p}$ and a low precession rate $\boldsymbol{\Omega}$ about the y-axis, the angular momentum is represented by the vector $\boldsymbol{H} = I\boldsymbol{p}$, where $I = I_{zz}$ is the moment of inertia of the rotor about the spin axis



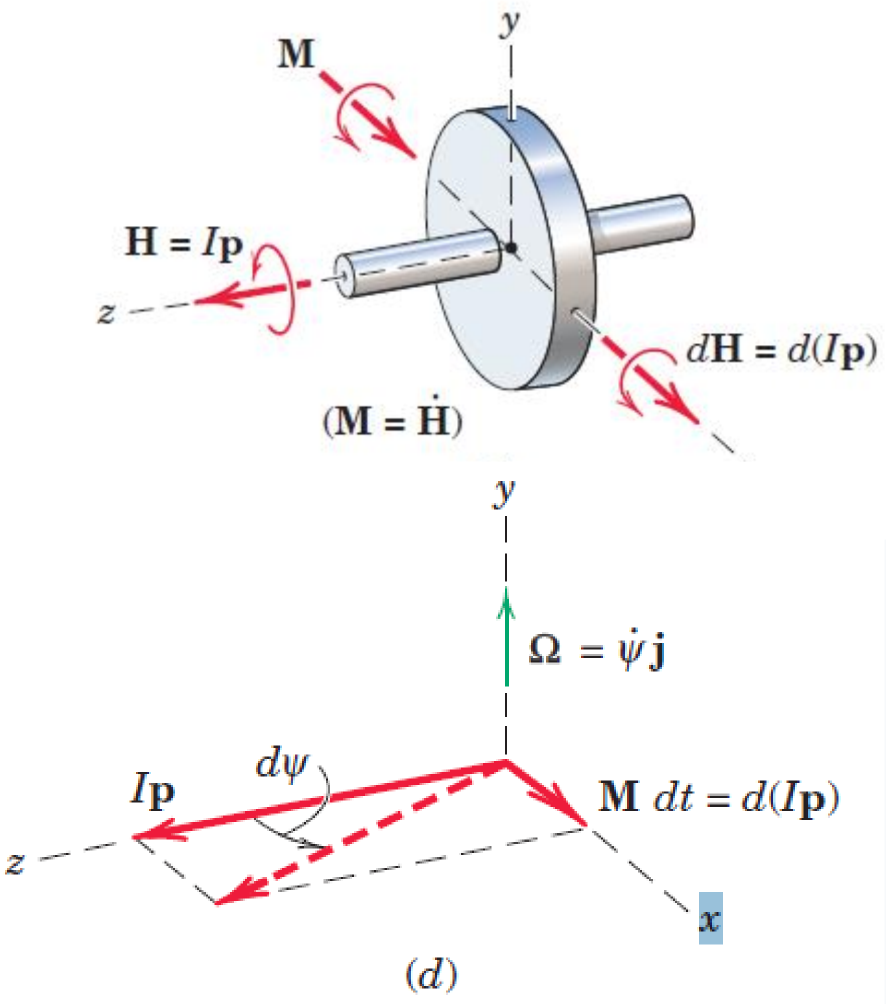
(c)



(d)

Gyroscopic Motion: Simplified Analysis

- Neglect the small component of angular momentum about the *y*-axis which accompanies the slow precession.
- The application of the couple **M** normal to **H** causes a change $d\mathbf{H}=d(I\mathbf{p})$ in the angular momentum
- $d\mathbf{H}$, and thus $d\mathbf{p}$, is a vector in the direction of the couple **M** and $\mathbf{M}\,dt=d\mathbf{H}$
- Just as the change in the linear-momentum vector of the particle is in the direction of the applied force, so is the change in the angular-momentum vector of the gyro in the direction of the couple
- Thus, we see that the vectors **M**, **H**, and $d\mathbf{H}$ are analogous to the vectors **F**, **G**, and $d\mathbf{G}$. With this insight, it is no longer strange to see the rotation vector undergo a change which is in the direction of **M**, thereby causing the axis of the rotor to precess about the *y*-axis.



Gyroscopic Motion: Simplified Analysis

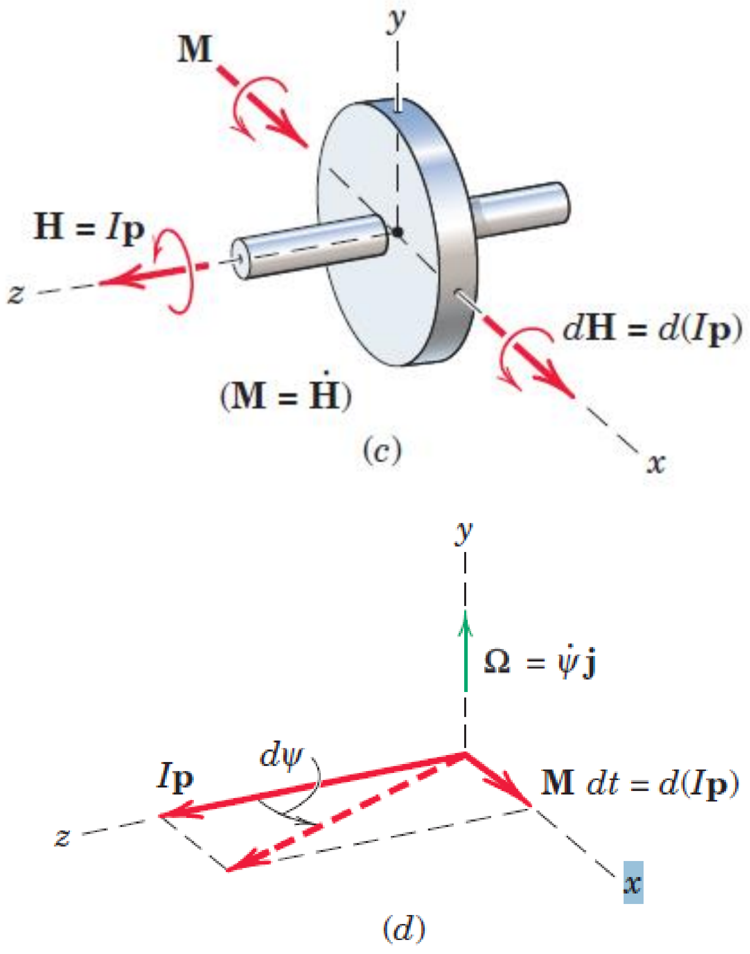
- During time dt the angular-momentum vector Ip has swung through the angle $d\psi$ so that in the limit with $\tan d\psi = d\psi$

$$d\psi = \frac{\mathbf{M}\,dt}{I\mathbf{p}}$$

$$\mathbf{M} = I\mathbf{p}\frac{d\psi}{dt}$$

$$\Omega = \frac{d\psi}{dt}$$

$$\mathbf{M} = I\mathbf{p}\Omega$$



M, **Ω**, and **p** as vectors are mutually perpendicular, and that their vector relationship may be represented by writing the equation in the cross-product form

$$\mathbf{M} = I\boldsymbol{\Omega} \times \mathbf{p}$$

Analogous to the foregoing relation $\mathbf{F} = m\boldsymbol{\omega} \times \mathbf{v}$ for the curvilinear motion of a particle

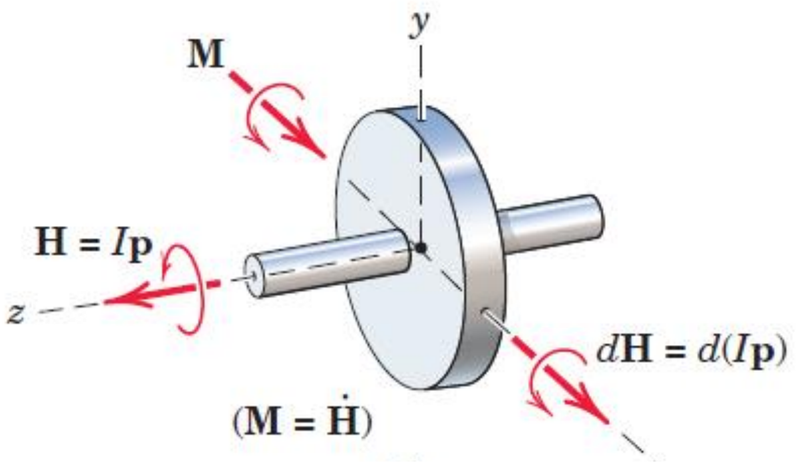
Gyroscopic Motion: Simplified Analysis

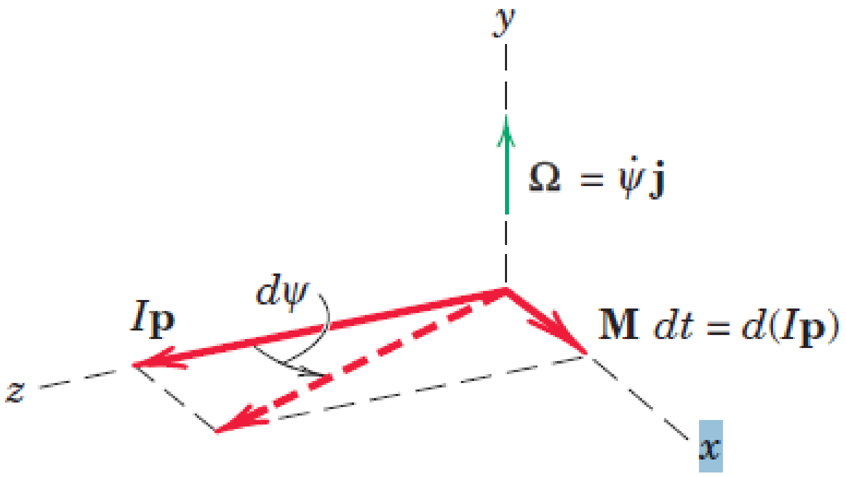
$$\mathbf{M} = I\mathbf{p}\Omega$$

M, **Ω**, and **p** as vectors are mutually perpendicular, and that their vector relationship may be represented by writing the equation in the cross-product form

$$\mathbf{M} = I\mathbf{\Omega} \times \mathbf{p}$$

- Equations apply to moments taken about the mass center or about a fixed point on the axis of rotation

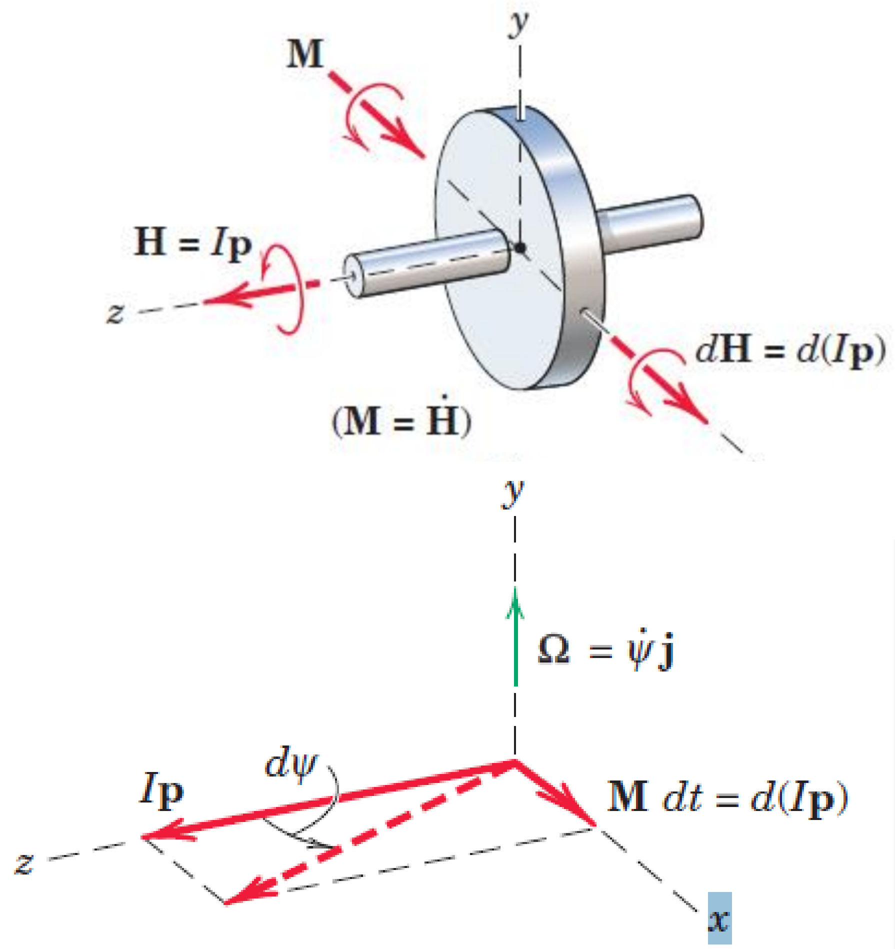




(d)

Gyroscopic Motion: Simplified Analysis

- The correct spatial relationship among the three vectors may be remembered from the fact that $d\mathbf{H}$, and thus $d\mathbf{p}$, is in the direction of **M**, which establishes the correct sense for the precession $\boldsymbol{\Omega}$. Therefore, the spin vector **p** always tends to rotate toward the torque vector **M**.



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Gyroscopic Motion: Simplified Analysis

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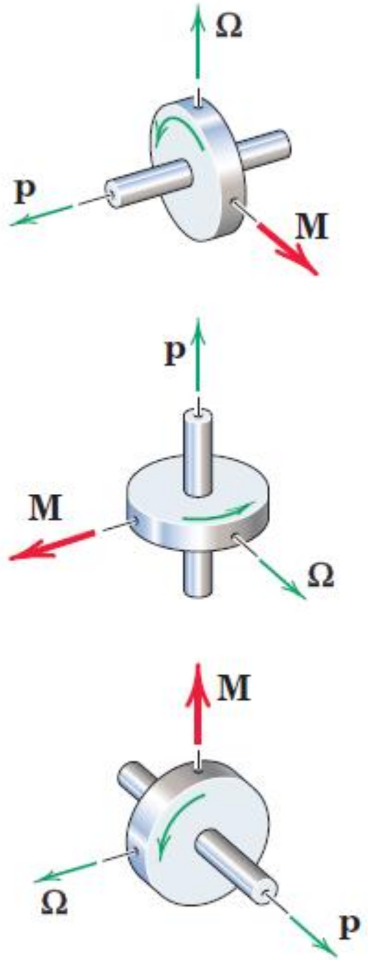
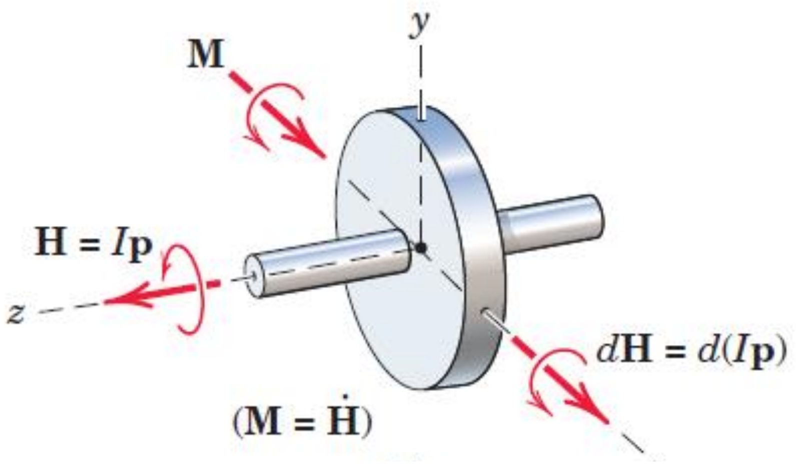
- Figure represents three orientations of the three vectors which are consistent with their correct order. Unless we establish this order correctly in a given problem, we are likely to arrive at a conclusion directly opposite to the correct one.

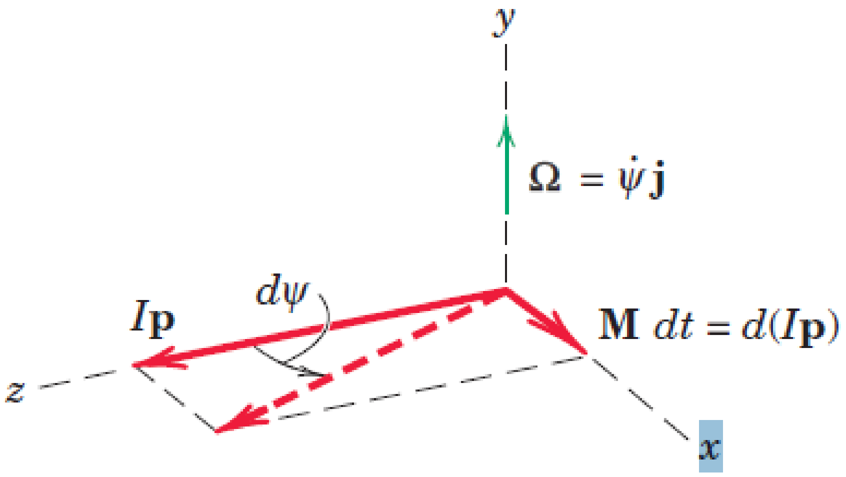


Gyroscopic Motion: Simplified Analysis

- $\boldsymbol{F} = m\boldsymbol{a}$ and $\boldsymbol{M} = I\boldsymbol{\alpha}$ equations of motion
  The couple **M** represents the couple due to all forces acting on the rotor, as disclosed by a correct free-body diagram of the rotor

- Also note that, when a rotor is forced to precess, as occurs with the turbine in a ship which is executing a turn, the motion will generate a gyroscopic couple **M** which obeys $\mathbf{M} = \boldsymbol{I\Omega} \times \mathbf{p}$ in both magnitude and sense

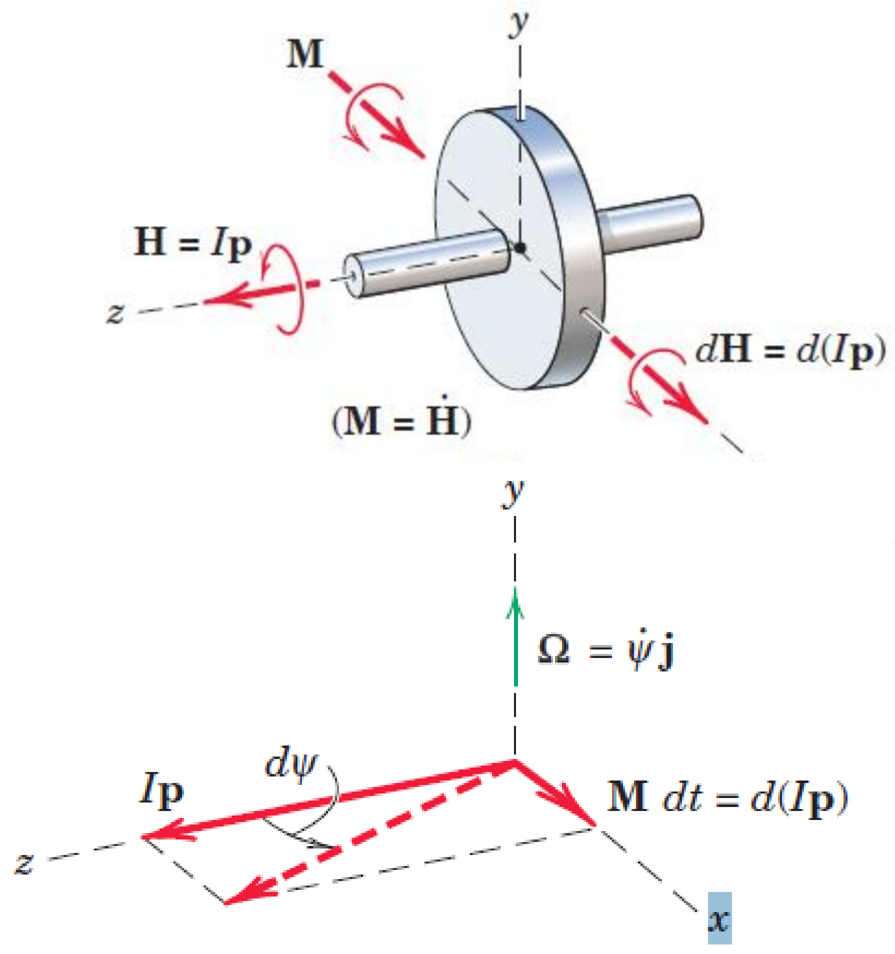




(d)

Gyroscopic Motion: Simplified Analysis

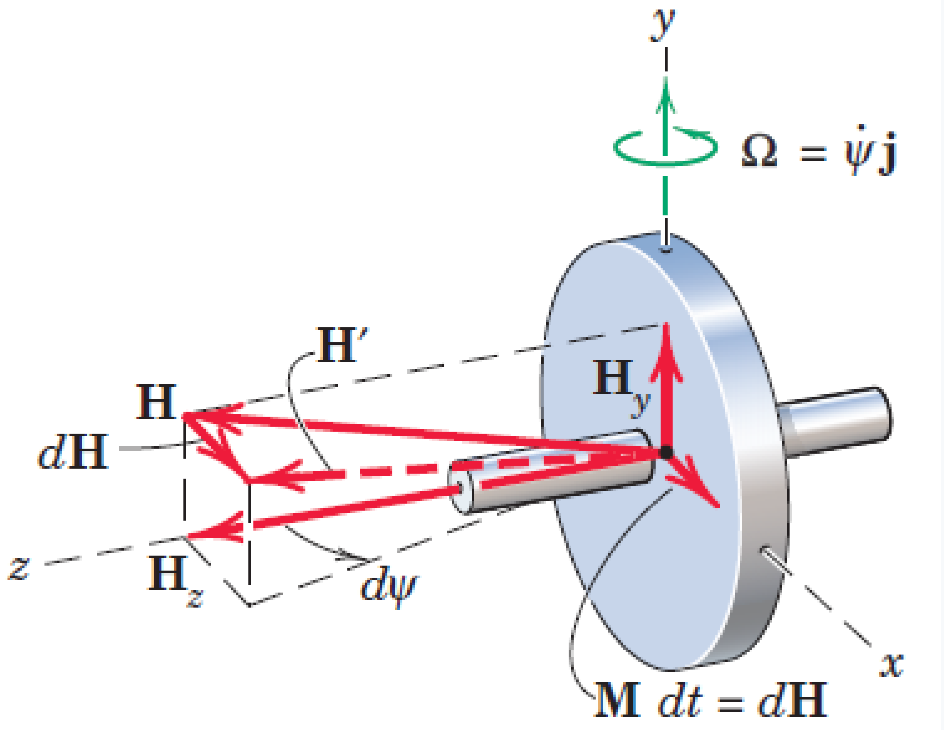
- In the foregoing discussion of gyroscopic motion, it was assumed that the spin was large and the precession was small. Although we can see from $\mathbf{M} = I\mathbf{p}\Omega$ that for given values of I and M, the precession $\boldsymbol{\Omega}$ must be small if p is large, **let us now examine the influence of $\boldsymbol{\Omega}$ on the momentum relations**. Again, we restrict our attention to steady precession, where has $\boldsymbol{\Omega}$ a constant magnitude.



(d)

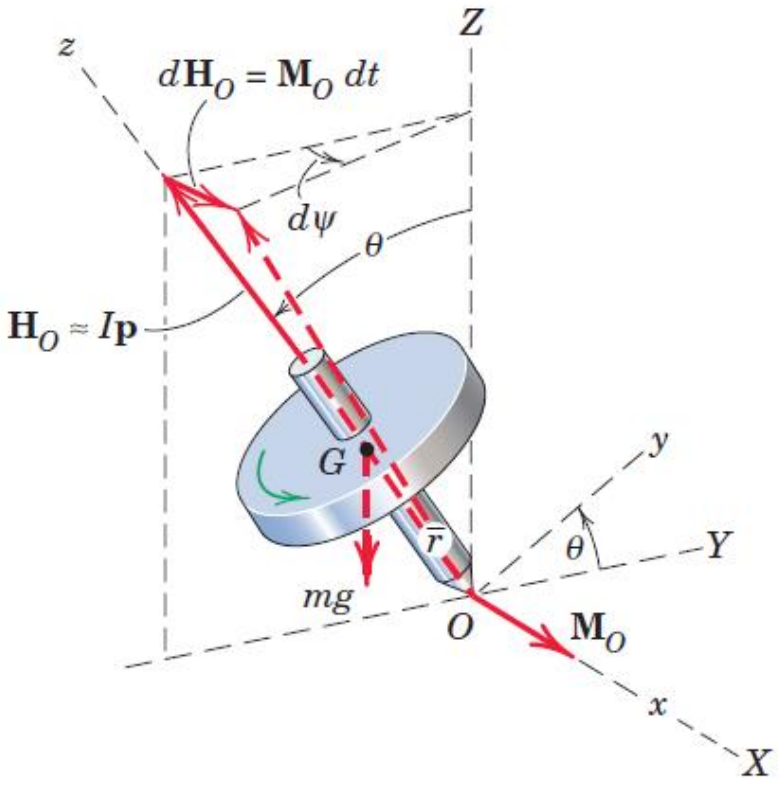
Gyroscopic Motion: Simplified Analysis

- Figure shows spinning rotor
- Because it has a moment of inertia about the y-axis and an angular velocity of precession about this axis, there will be an additional component of angular momentum about the y-axis. Thus, we have the two components $H_z = Ip$ and $H_y = I_o\Omega$,where I_o stands for I_{yy} and I stands for I_{zz}.
- The total angular momentum is H as shown in figure. The change in H remains $d\mathbf{H} = \mathbf{M}\,dt$ as previously, and the precession during time dt is the angle $d\psi = \mathbf{M}\,dt/\mathbf{H}_z = \mathbf{M}\,dt/Ip$ as before.
- Thus, $\mathbf{M} = \mathbf{I\Omega p}$ **is still valid and for steady precession is an exact description of the motion as long as the spin axis is perpendicular to the axis around which precession occurs**



Gyroscopic Motion: Simplified Analysis

- Consider now the steady precession of a symmetrical top spinning about its axis with a high angular velocity p and supported at its point O
- Here the spin axis makes an angle θ with the vertical z-axis around which precession occurs. Again, we will neglect the small angular-momentum component due to the precession and consider **H** equal to $I\mathbf{p}$, the angular momentum about the axis of the top associated with the spin only.
- The moment about O is due to the weight mg and is $mg\bar{r}\sin\theta$, where $\bar{r}$ is the distance from O to the mass center G.



Gyroscopic Motion: Simplified Analysis

- From the diagram, we see that the angular-momentum vector $\mathbf{H}_o$ has a change $d\mathbf{H}_o = \mathbf{M}_o dt$ in the direction of $\mathbf{M}_o$ during time dt and that θ is unchanged.
- The increment in precessional angle around the Z-axis is

$$d\psi = \frac{M_o dt}{Ip\ \sin\theta}$$

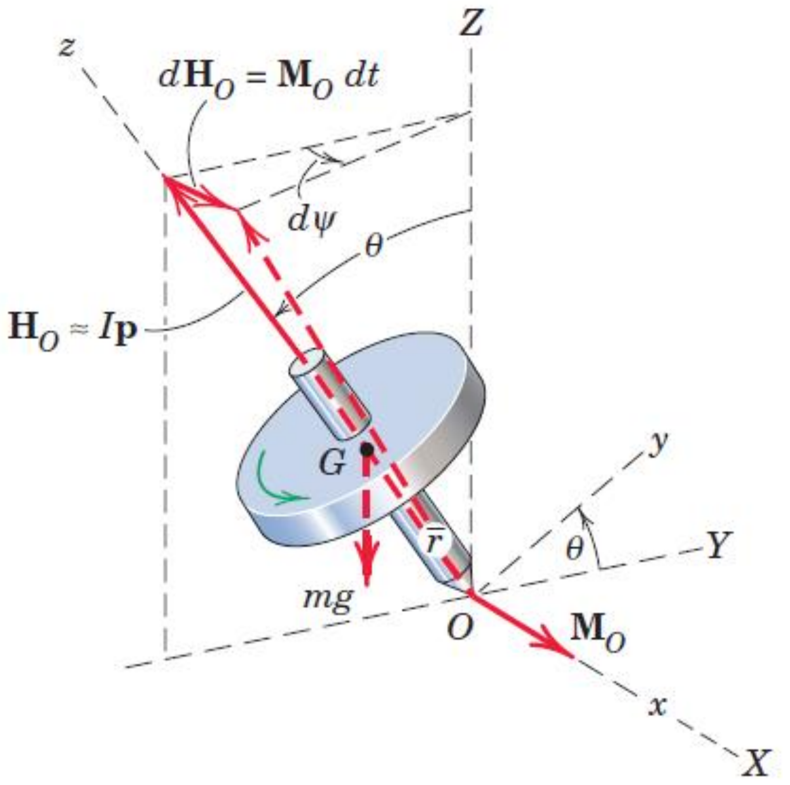
Substituting the values $M_o = mg\bar{r}\sin\theta$ and $\Omega = \frac{d\psi}{dt}$ gives

$$mg\bar{r} = Ip\Omega$$

which is independent of θ.

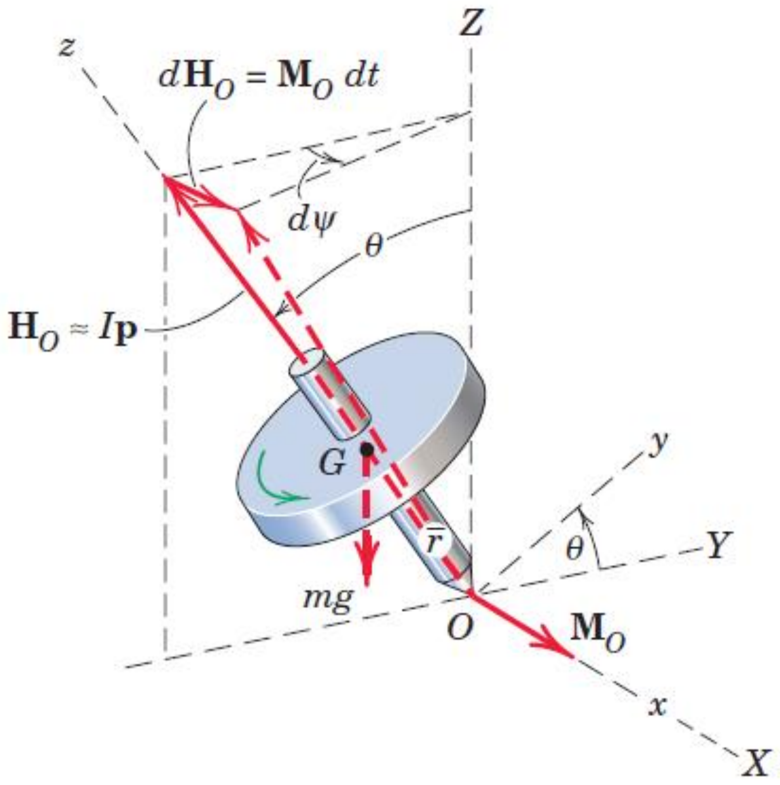
Introducing the radius of gyration so that $I = mk^2$ and solving for the precessional velocity give

$$\Omega = \frac{g\bar{r}}{k^2 p}$$



Gyroscopic Motion: Simplified Analysis

- Unlike $\mathbf{M} = I\mathbf{p}\boldsymbol{\Omega}$, which is an exact description for the rotor with precession confined to the x-z plane, $\Omega = \frac{g\bar{r}}{k^2 p}$ is an approximation based on the assumption that the angular momentum associated with Ω is negligible compared with that associated with p. We will see the amount of the error associated with this approximation when we reconsider steady-state precession later.
- On the basis of analysis (mentioned in Meriam and Kraige), the top will have a steady precession at the constant angle θ only if it is set in motion with a value of Ω which satisfies $\Omega = \frac{g\bar{r}}{k^2 p}$. When these conditions are not met, the precession becomes unsteady, and θ may oscillate with an amplitude which increases as the spin velocity decreases. The corresponding rise and fall of the rotation axis is called **nutation**.



Momentum Equations

$$\sum \boldsymbol{M} = \left(\frac{\mathrm{d}\boldsymbol{H}}{\mathrm{d}t}\right)_{xyz} + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H_y}\Omega_\mathrm{z} + \mathrm{H_z}\Omega_\mathrm{y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H_z}\Omega_\mathrm{x} + \mathrm{H_x}\Omega_\mathrm{z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H_x}\Omega_\mathrm{y} + \mathrm{H_y}\Omega_\mathrm{x}\right)\mathrm{k}$$

Most general form of the moment equation about a fixed point *O* or about the mass center *G*

Ω's are the angular velocity components of rotation of the reference axes

In the *H* components of a rigid body, ω the angular velocity of the body is used

Momentum Equations

$$\sum \boldsymbol{M} = \left(\dot{H}_x \mathbf{i} + \dot{H}_y \mathbf{j} + \dot{H}_z \mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

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Most general form of the moment equation about a fixed point *O* or about the mass center *G*.
Ω's are the angular velocity components of rotation of the reference axes
In the *H* components of a rigid body, ω the angular velocity of the body is used

$$H_x = \left[(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z\right]$$

$$H_y = \left[-I_{xy}\omega_x + \left(I_{yy}\right)\omega_y - I_{yz}\omega_z\right]$$

$$H_z = \left[-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z\right]$$

Gyroscopic Motion: Detailed Analysis

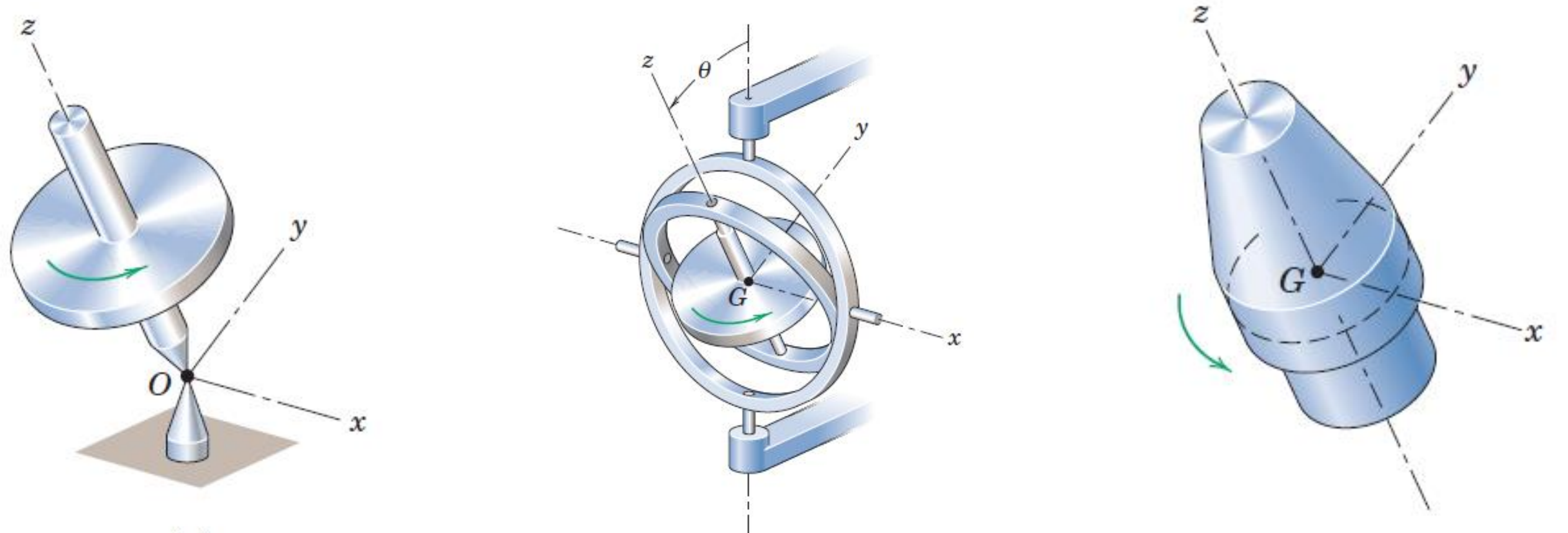
- Make direct use of Momentum Equations equation, which is the general angular momentum equation for a rigid body, by applying it to a body spinning about its axis of rotational symmetry.

- This equation is valid for rotation about a fixed point or for rotation about the mass center.

- A spinning top, the rotor of a gyroscope, and a spacecraft are examples of bodies whose motions can be described by the equations for rotation about a point.

Gyroscopic Motion: Detailed Analysis

- **A spinning top, the rotor of a gyroscope, and a spacecraft** are examples of bodies whose motions can be described by the equations for rotation about a point.



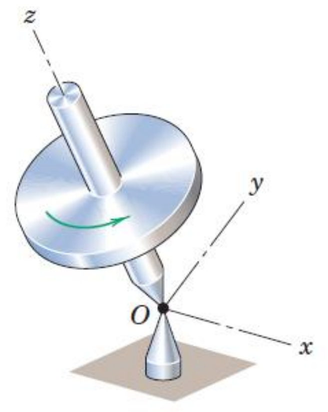
(a) (b) (c)

Gyroscopic Motion: Detailed Analysis

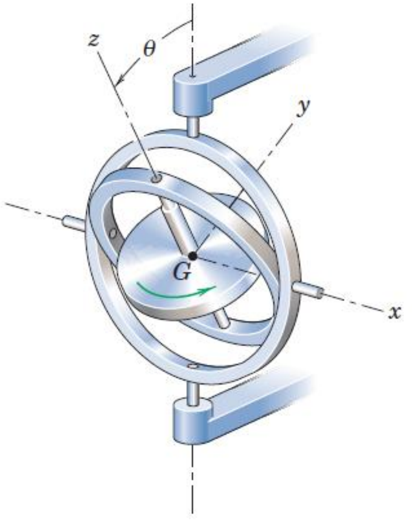
- The general moment equations for this class of problems are fairly complex, and their complete solutions involve the use of elliptic integrals and somewhat lengthy computations.
- However, a large fraction of engineering problems where the motion is one of rotation about a point involves the **steady precession of bodies of revolution which are spinning about their axes of symmetry**. These conditions greatly simplify the equations and thus facilitate their solution.

Gyroscopic Motion: Detailed Analysis

- Consider a body with axial symmetry rotating about a fixed point O on its axis, which is taken to be the z-direction. With O as origin, **the x- and y-axes automatically become principal axes of inertia along with the z-axis**.
- This same description may be used for the rotation of a similar symmetrical body about its center of mass G, which is taken as the origin of coordinates as shown with the gimbaled gyroscope rotor. Again, the x- and y-axes are principal axes of inertia for point G.
- The same description may also be used to represent the rotation about the mass center of an axially symmetric body in space, such as the spacecraft. In each case, we note that, regardless of the rotation of the axes or of the body relative to the axes (spin about the z-axis), the moments of inertia about the x- and y-axes remain constant with time.
- The principal moments of inertia are again designated $I_{zz} = I$ and $I_{xx} = I_{yy} = I_o$. The products of inertia are, of course, zero.



(a)



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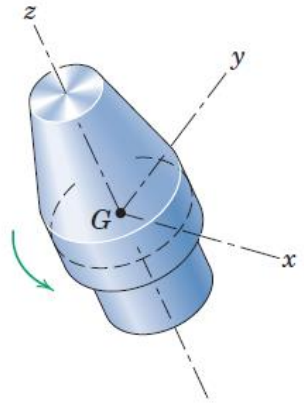
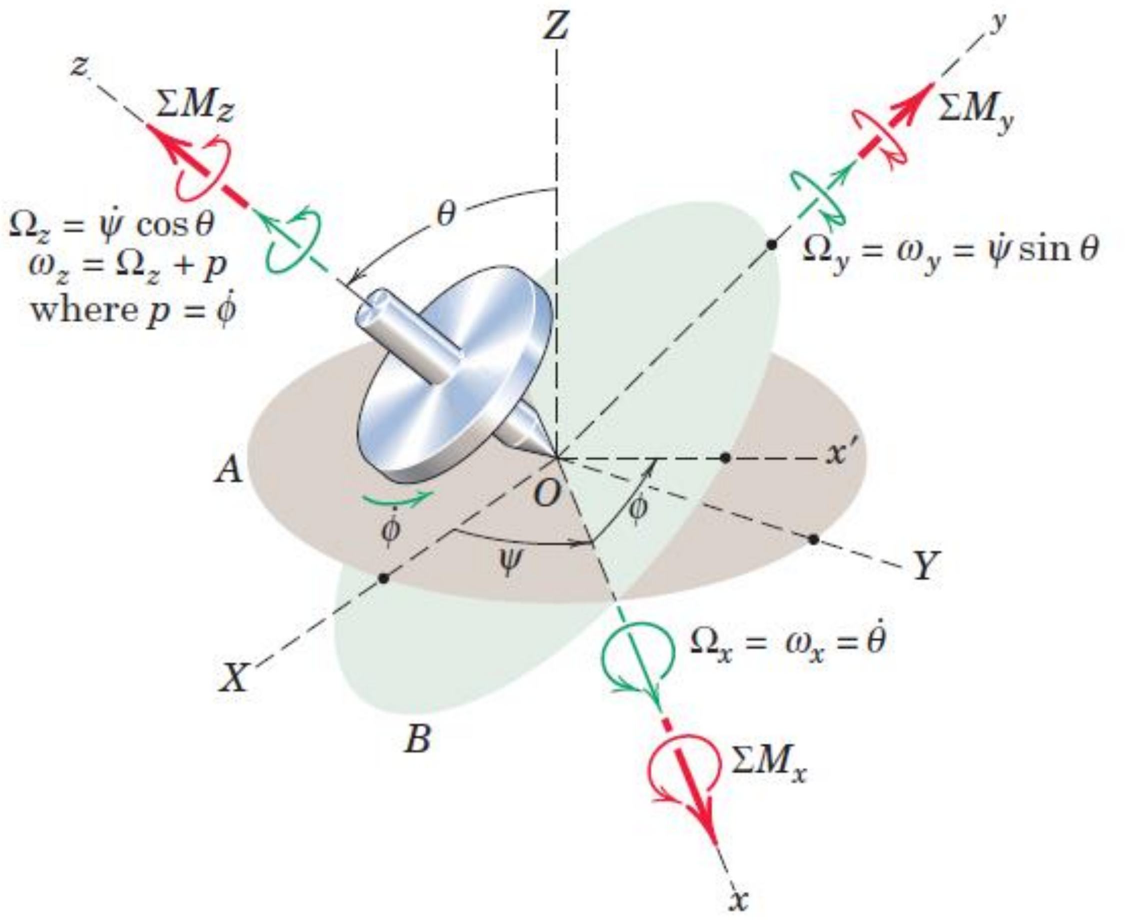


(c)

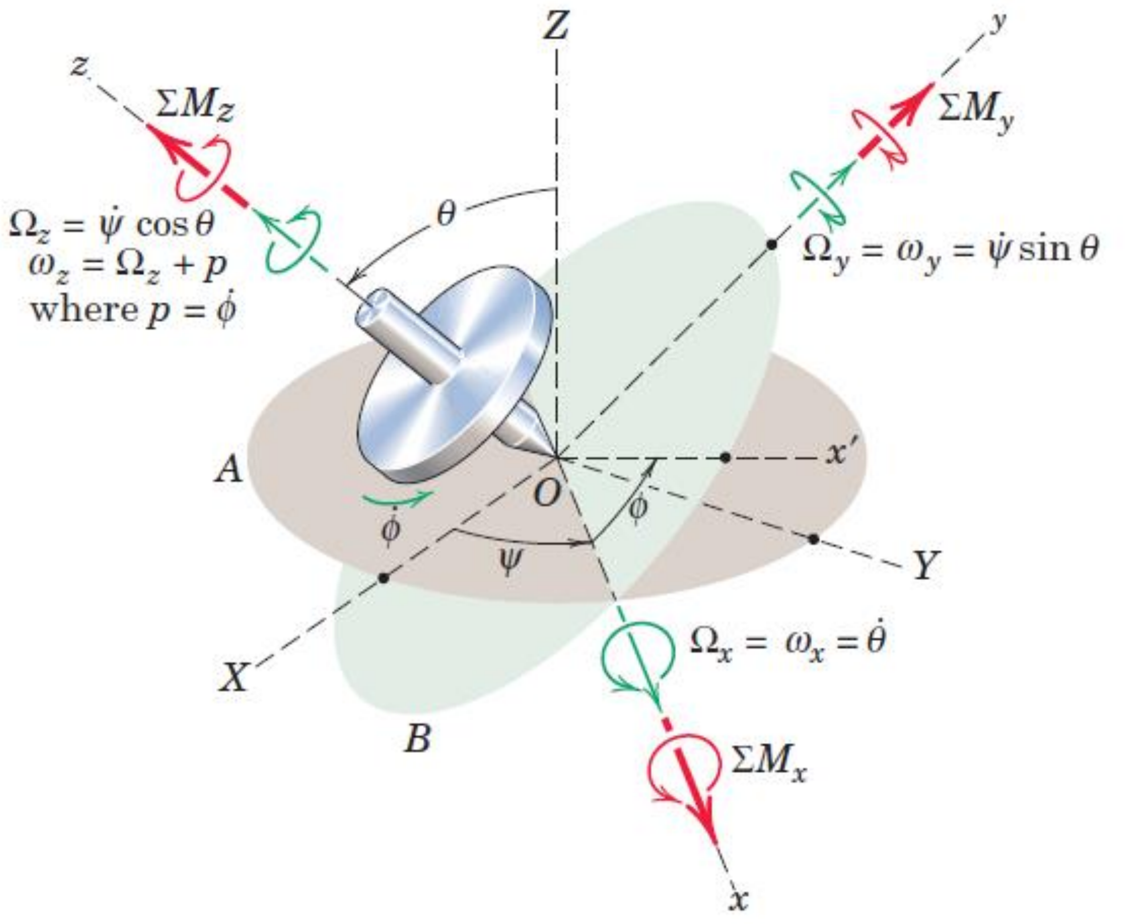
Gyroscopic Motion: Detailed Analysis

- The axes X-Y-Z are fixed in space, and plane A contains the X-Y axes and the fixed point O on the rotor axis.
- Plane B contains point O and is always normal to the rotor axis.
- Angle θ measures the inclination of the rotor axis from the vertical Z-axis and is also a measure of the angle between planes A and B.
- The intersection of the two planes is the *x-axis*, which is located by the angle ψ from the X-axis. The y-axis lies in plane B, and the z-axis coincides with the rotor axis. The angles θ and ψ completely specify the position of the rotor axis.



Gyroscopic Motion: Detailed Analysis

- The angular displacement of the rotor with respect to axes x-y-z is specified by the angle ϕ measured from the x-axis to the x'-axis, which is attached to the rotor. The spin velocity becomes $p = \dot{\phi}$



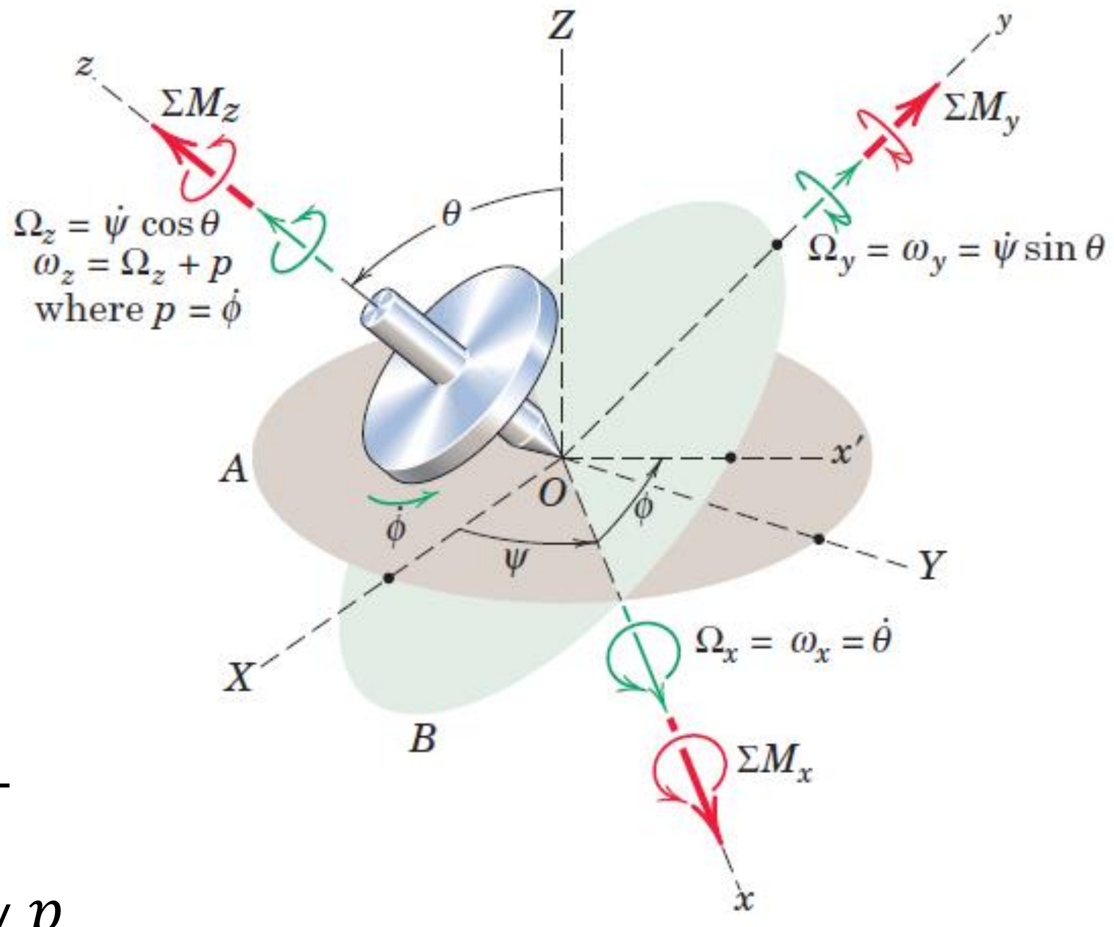
Gyroscopic Motion: Detailed Analysis

- The components of the angular velocity ω of the rotor and the angular velocity Ω of the axes x-y-z from

$$\Omega_x = \dot{\theta} \qquad \omega_x = \dot{\theta}$$

$$\Omega_y = \dot{\psi} \sin\theta \qquad \omega_y = \dot{\psi} \sin\theta$$

$$\Omega_z = \dot{\psi} \cos\theta \qquad \omega_z = \dot{\psi} \cos\theta + p$$



Note: Axes and the body have identical x- and y-components of angular velocity, but that the z-components differ by the relative angular velocity p

Gyroscopic Motion: Detailed Analysis

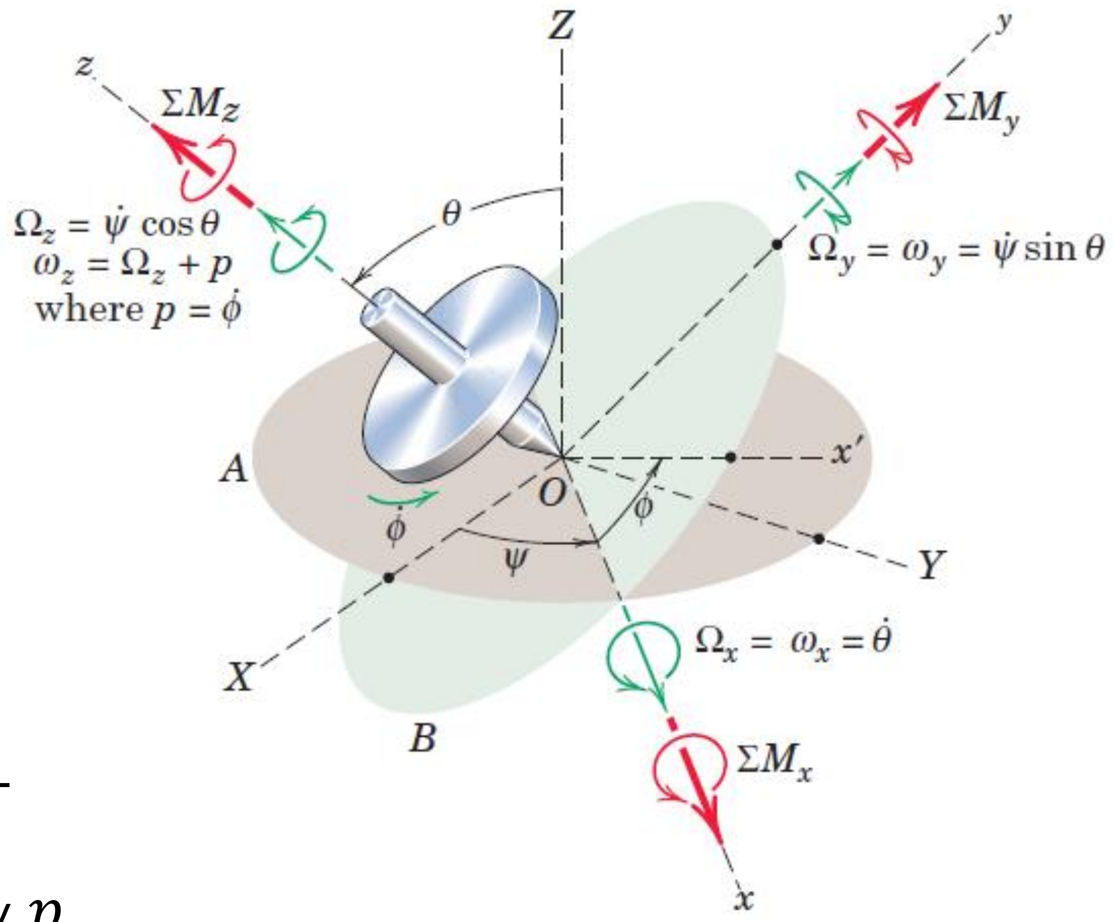
- Angular momentum equations

$$H_x = I_{xx}\omega_x = I_o\dot{\theta}$$

$$H_y = I_{yy}\omega_y = I_o\dot{\psi}\sin\theta$$

$$H_z = I_{zz}\omega_z = I(\dot{\psi}\ \cos\theta + p)$$

Note: Axes and the body have identical x- and y-components of angular velocity, but that the z-components differ by the relative angular velocity p



Momentum Equations

$$\sum \boldsymbol{M} = \left(\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}\right) + \boldsymbol{\Omega} \times \mathbf{H}$$

$$\sum \mathrm{M} = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H}_\mathrm{y}\Omega_\mathrm{z} + \mathrm{H}_\mathrm{z}\Omega_\mathrm{y}\right)\mathrm{i} + \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H}_\mathrm{z}\Omega_\mathrm{x} + \mathrm{H}_\mathrm{x}\Omega_\mathrm{z}\right)\mathrm{j} + \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H}_\mathrm{x}\Omega_\mathrm{y} + \mathrm{H}_\mathrm{y}\Omega_\mathrm{x}\right)\mathrm{k}$$

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$$H_z = I_{zz}\omega_z = I\left(\dot{\psi}\ \cos\theta + p\right)$$

$$\sum M_x = \left(\dot{\mathrm{H}}_\mathrm{x} - \mathrm{H}_\mathrm{y}\Omega_\mathrm{z} + \mathrm{H}_\mathrm{z}\Omega_\mathrm{y}\right)$$

$$\sum M_y = \left(\dot{\mathrm{H}}_\mathrm{y} - \mathrm{H}_\mathrm{z}\Omega_\mathrm{x} + \mathrm{H}_\mathrm{x}\Omega_\mathrm{z}\right)$$

$$\sum M_z = \left(\dot{\mathrm{H}}_\mathrm{z} - \mathrm{H}_\mathrm{x}\Omega_\mathrm{y} + \mathrm{H}_\mathrm{y}\Omega_\mathrm{x}\right)\mathrm{k}$$

Momentum Equations

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$$H_x = I_{xx}\omega_x = I_o\dot{\theta}$$

$$\boxed{\sum M_x = I_o\left(\ddot{\theta} - \dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}\left(\dot{\psi}\ \cos\theta + p\right)\sin\theta}$$

$$H_y = I_{yy}\omega_y = I_o\dot{\psi}\sin\theta$$

$$H_z = I_{zz}\omega_z = I\left(\dot{\psi}\ \cos\theta + p\right)$$

$$\boxed{\sum M_y = \left(I_o\ddot{\psi}\sin\theta + 2I_o\dot{\theta}\dot{\psi}\cos\theta - I\left(\dot{\psi}\ \cos\theta + p\right)\dot{\theta}\right)}$$

$$\boxed{\sum M_z = \left(I\left(\ddot{\psi}\ \cos\theta - \dot{\psi}\ \dot{\theta}\sin\theta + \dot{p}\right) - I_o\dot{\theta}\dot{\psi}\sin\theta + I_o\dot{\theta}\dot{\psi}\sin\theta\right) = I\left(\ddot{\psi}\ \cos\theta - \dot{\psi}\ \dot{\theta}\sin\theta + \dot{p}\right)}$$

Momentum Equations

$$\boxed{\sum M_x = I_o\left(\ddot{\theta} - \dot{\psi}\sin\theta\,(\dot{\psi}\ \cos\theta)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta}$$

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- Above equations are the general equations of rotation of a symmetrical body about either a fixed point O or the mass center G.
- In a given problem, the solution to the equations will depend on the moment sums applied to the body about the three coordinate axes.

Steady State Precession

- We now examine the conditions under which the rotor precesses at a steady rate $\dot{\psi}$ at a constant angle θ and with constant spin velocity p.

$$\dot{\psi} = constant \qquad \ddot{\psi} = 0$$

$$\theta = constant \qquad \dot{\theta} = \ddot{\theta} = 0$$

$$p = constant \qquad \dot{p} = 0$$

$$\boxed{\sum M_x = I_o\left(-\dot{\psi}\sin\theta\,(\dot{\psi}\ \cos\theta)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta}$$

$$\boxed{\sum M_y = (I_o\ddot{\psi}\sin\theta + 2I_o\dot{\theta}\dot{\psi}\cos\theta - I(\dot{\psi}\ \cos\theta + p)\dot{\theta}) = 0}$$

$$\boxed{\sum M_z = I(\ddot{\psi}\ \cos\theta - \dot{\psi}\ \dot{\theta}\sin\theta + \dot{p}) = 0}$$

Steady State Precession

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$$\boxed{\sum M_y = (I_o\ddot{\psi}\sin\theta + 2I_o\dot{\theta}\dot{\psi}\cos\theta - I(\dot{\psi}\ \cos\theta + p)\dot{\theta})=0}$$

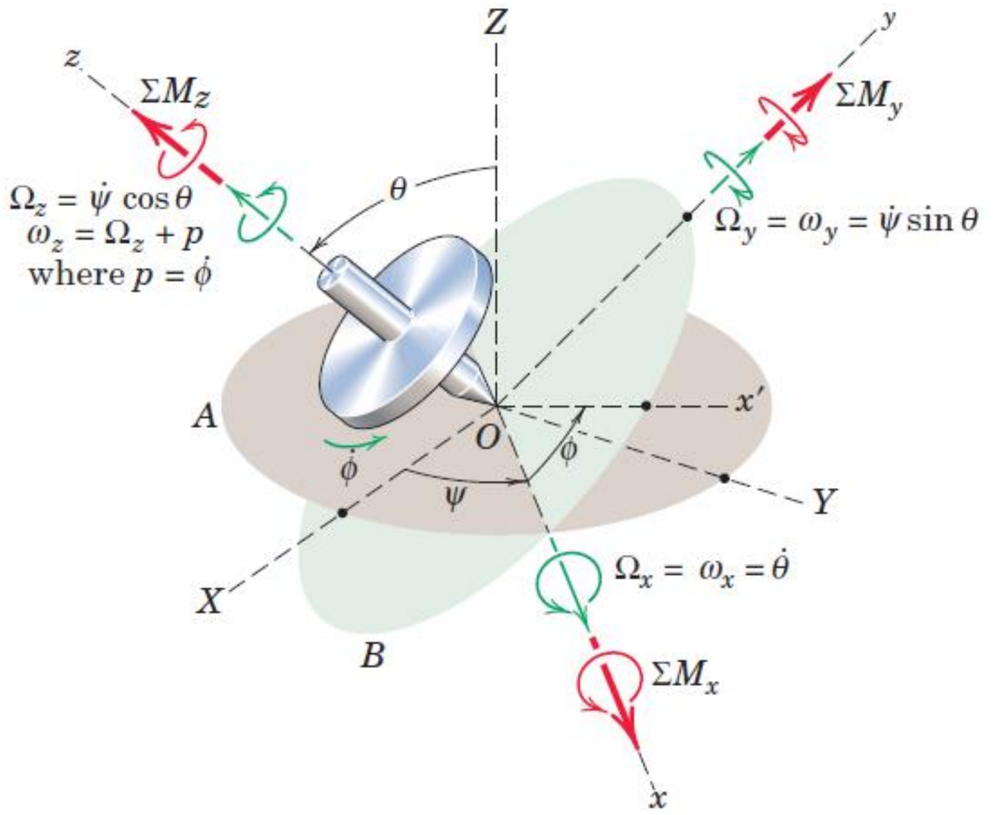
$$\boxed{\sum M_z = I(\ddot{\psi}\ \cos\theta - \dot{\psi}\ \dot{\theta}\sin\theta + \dot{p}) = 0}$$

- Required moment acting on the rotor about O (or about G) must be in the x-direction since the y- and z-components are zero.
- With the constant values of θ, $\dot{\psi}$ and p, the moment is constant in magnitude.
- Moment axis is perpendicular to the plane defined by the precession axis (Z-axis) and the spin axis (z-axis)

Steady State Precession

$$\sum M_x = I_o\left(-\dot{\psi}\sin\theta\,(\dot{\psi}\ \cos\theta)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta$$

$$\sum M_y = 0 \qquad \sum M_z = 0$$



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Steady State Precession

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$$\sum M_y = 0$$

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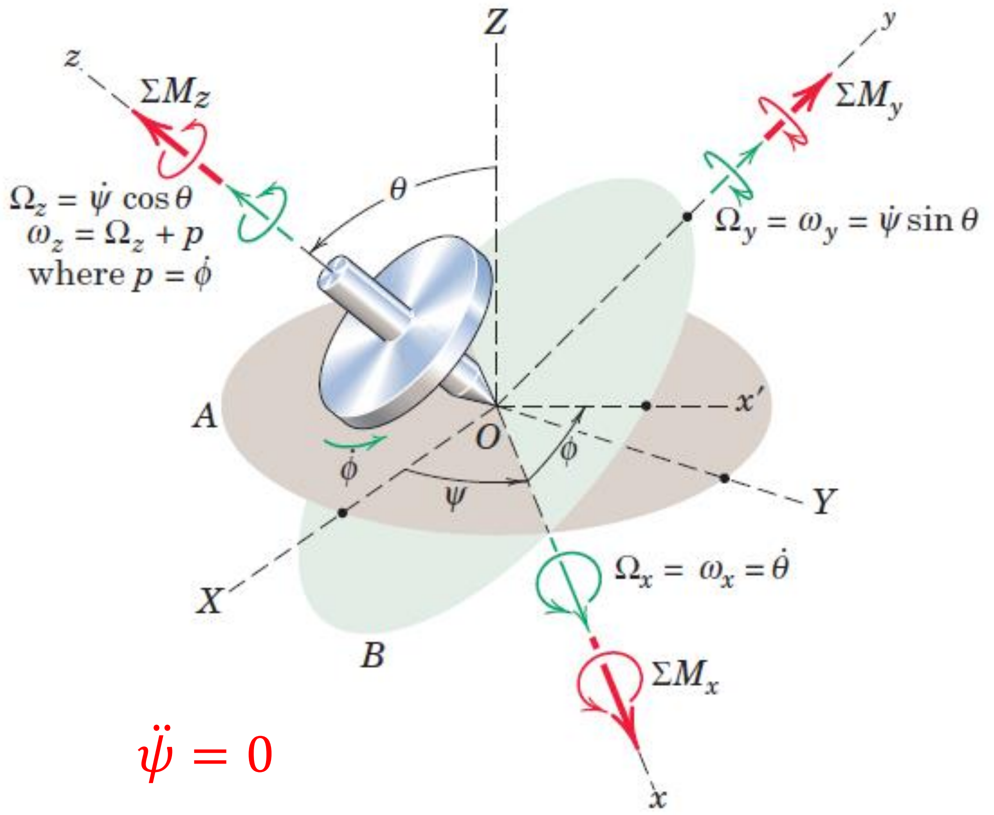


$$\sum \boldsymbol{M} = (\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}) + \boldsymbol{\Omega}\times\mathbf{H}$$

$$H_x = I_{xx}\omega_x = I_o\dot{\theta}$$

$$H_y = I_{yy}\omega_y = I_o\dot{\psi}\sin\theta$$

$$H_z = I_{zz}\omega_z = I(\dot{\psi}\ \cos\theta + p)$$

$\dot{\psi} = constant$ $\qquad \ddot{\psi} = 0$

$\theta = constant$ $\qquad \dot{\theta} = \ddot{\theta} = 0$

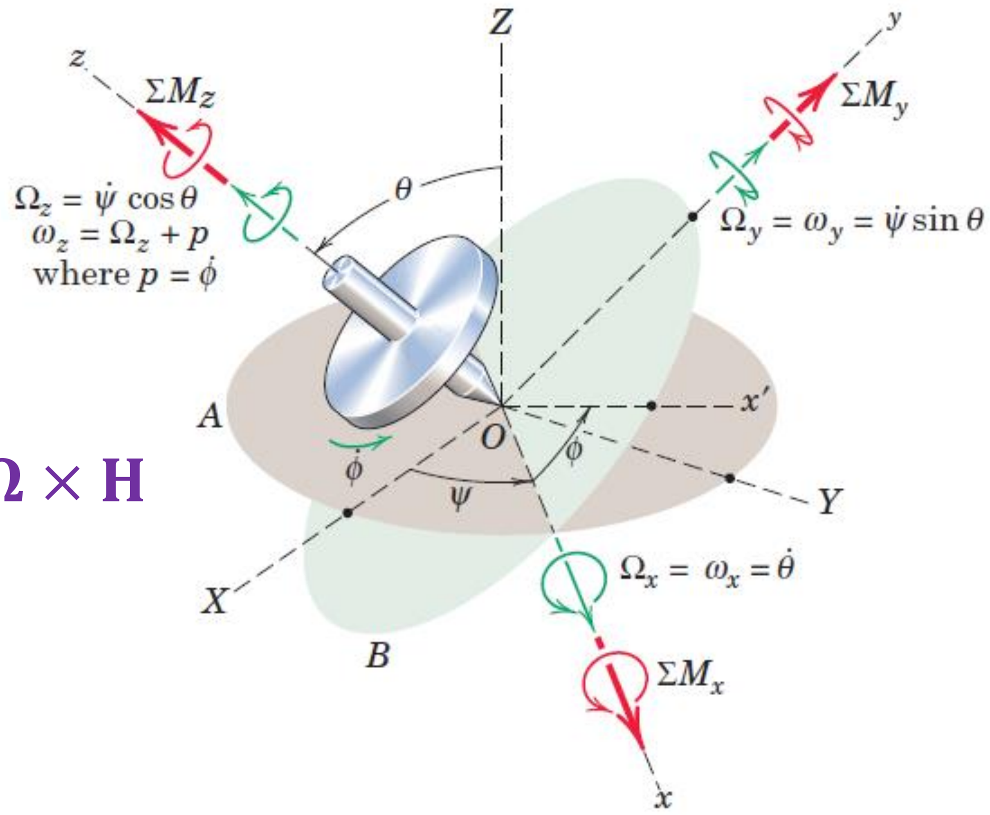
$p = constant$ $\qquad \dot{p} = 0$

Steady State Precession

$$\boxed{\sum M_x = I_o\left(-\dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta}$$

$$\boxed{\sum M_y = 0} \qquad \boxed{\sum M_z = 0}$$

$$\sum \boldsymbol{M} = (\dot{H}_x\mathbf{i} + \dot{H}_y\mathbf{j} + \dot{H}_z\mathbf{k}) + \boldsymbol{\Omega} \times \mathbf{H} = \mathbf{0} + \boldsymbol{\Omega} \times \mathbf{H}$$



$$H_x = I_{xx}\omega_x = I_o\dot{\theta} \qquad \Omega_x = \dot{\theta}$$

$$H_y = I_{yy}\omega_y = I_o\dot{\psi}\sin\theta \qquad \Omega_y = \dot{\psi}\sin\theta$$

$$H_z = I_{zz}\omega_z = I(\dot{\psi}\ \cos\theta + p) \qquad \Omega_z = \dot{\psi}\ \cos\theta$$

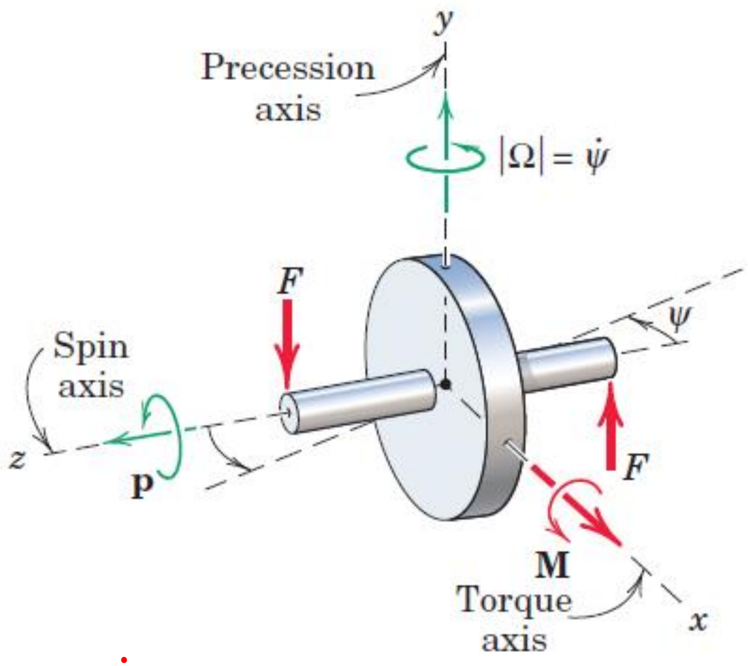
Steady State Precession

By far the most common engineering examples of gyroscopic motion occur when precession takes place about an axis which is normal to the rotor axis

$$\sum M_x = I_o\left(-\dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}\left(\dot{\psi}\ \cos\theta + p\right)\sin\theta$$

$$= I_o\left(-\dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}\left(\dot{\psi}\ \cos\theta + p\right)\sin\theta$$

$$= \dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)(I - I_o) + I\dot{\psi}\, p\sin\theta = I\dot{\psi}\, p = \ I\Omega\, p$$



$$H_x = I_{xx}\omega_x = I_o\dot{\theta} = I_o\dot{\theta}$$

$$H_y = I_{yy}\omega_y = I_o\dot{\psi}\sin\theta = I_o\dot{\psi}$$

$$H_z = I_{zz}\omega_z = I\left(\dot{\psi}\ \cos\theta + p\right) = Ip$$

$$\Omega_x = \dot{\theta} = 0$$

$$\Omega_y = \dot{\psi}\sin\theta = \dot{\psi}$$

$$\Omega_z = \dot{\psi}\ \cos\theta = 0$$

$$\dot{\psi} = constant$$

$$\theta = \frac{\pi}{2}$$

$$p = constant$$

Steady State Precession

By far the most common engineering examples of gyroscopic motion occur when precession takes place about an axis which is normal to the rotor axis

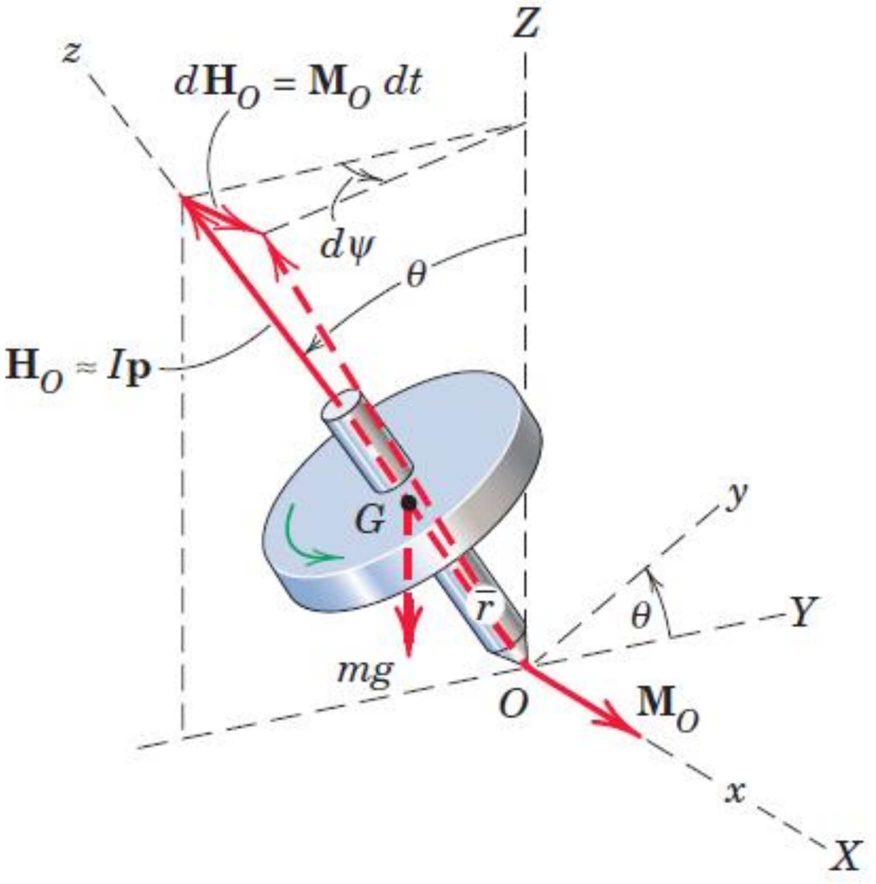
$\dot{\psi} = constant$ $\theta \neq \frac{\pi}{2}$ $p = constant$

$$\sum M_x = I_o\left(-\dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta$$

$$= I_o\left(-\dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)\right) + I\dot{\psi}(\dot{\psi}\ \cos\theta + p)\sin\theta$$

$$= \dot{\psi}\sin\theta\left(\dot{\psi}\ \cos\theta\right)(I - I_o) + I\dot{\psi}\, p\sin\theta = mg\bar{r}\sin\theta$$

$$\left(\dot{\psi}^2\cos\theta\right)(I - I_o) + I\dot{\psi}\, p = mg\bar{r}$$



Steady State Precession

By far the most common engineering examples of gyroscopic motion occur when precession takes place about an axis which is normal to the rotor axis

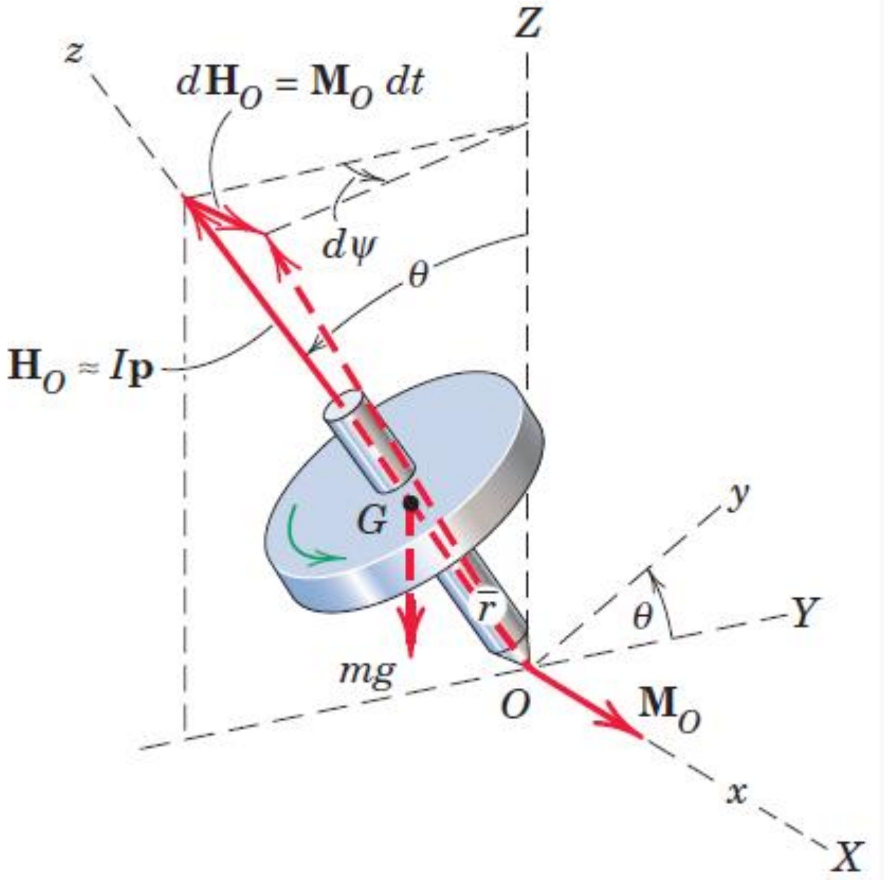
$\dot{\psi} = constant$ $\theta \neq \frac{\pi}{2}$ $p = constant$

$$(\dot{\psi}^2 \cos\theta)(I - I_o) + I\dot{\psi}\, p = mg\bar{r}$$

$\dot{\psi}$ is small if p is large,

$$I\dot{\psi}\, p = mg\bar{r} \qquad \dot{\psi} = \frac{mg\bar{r}}{Ip} \qquad \Omega = \frac{mg\bar{r}}{mk^2p} = \frac{g\bar{r}}{k^2p}$$

$$\Omega = \frac{g\bar{r}}{k^2p}$$

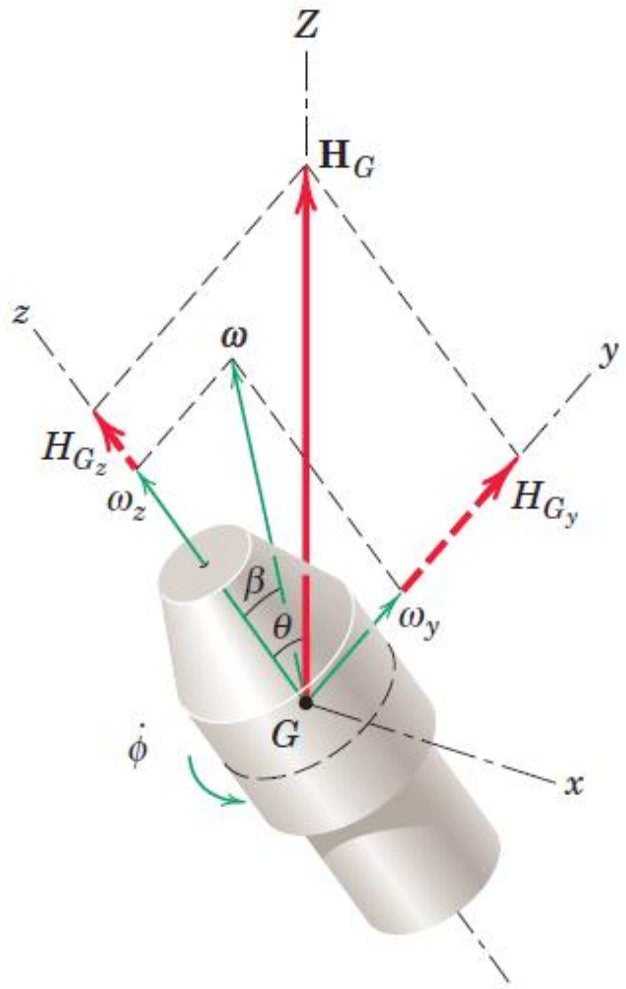


Steady State Precession with Zero Moment

Motion of a symmetrical rotor with **no external moment about its mass center**

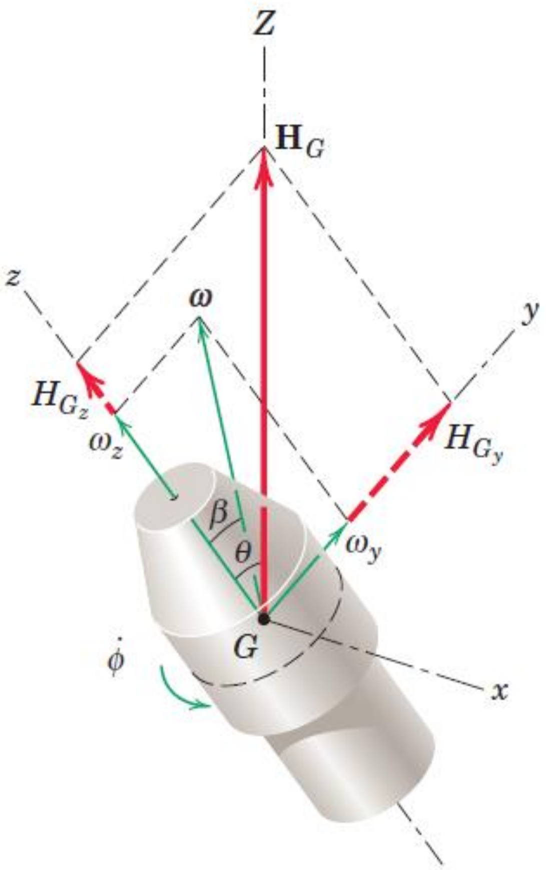
Such motion is encountered with spacecraft and projectiles which both spin and precess during flight

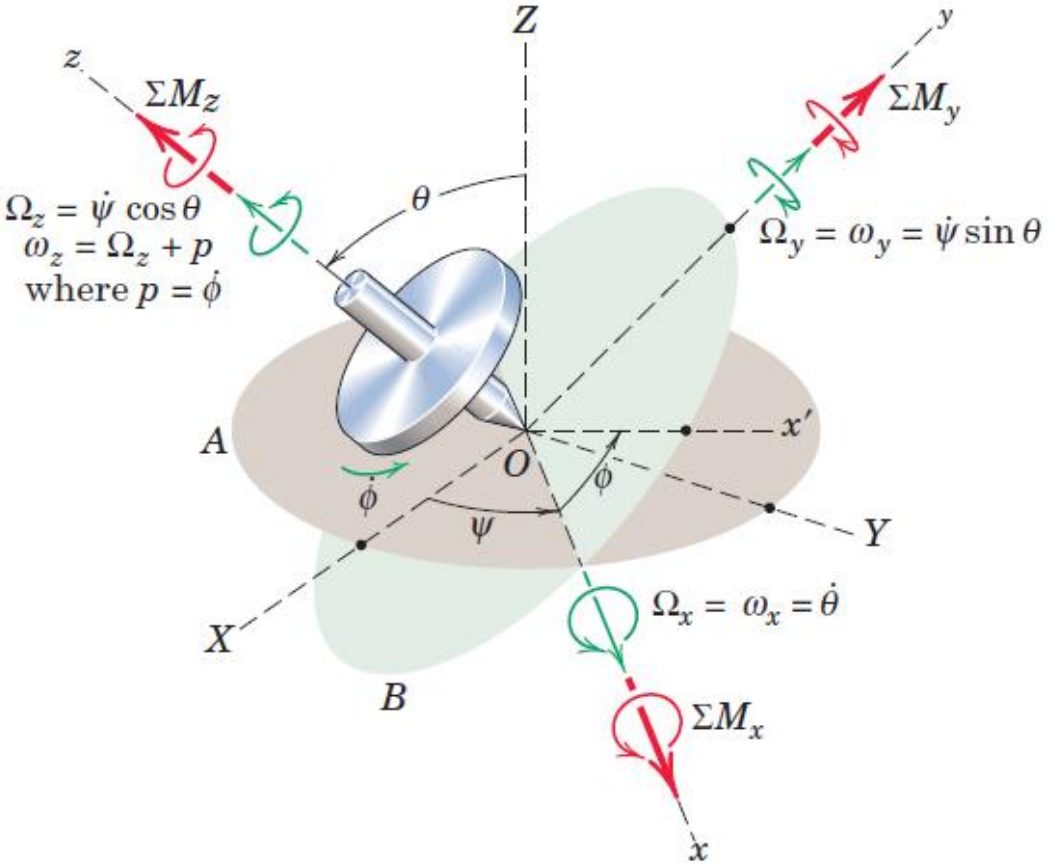
Z-axis, which has a fixed direction in space, is chosen to coincide with the direction of the angular momentum $\mathbf{H_G}$, which is constant since $\sum M_G = 0$



Steady State Precession with Zero Moment

The x-y-z axes are attached in the manner as shown in right figure





Steady State Precession with Zero Moment

Z-axis, which has a fixed direction in space, is chosen to coincide with the direction of the angular momentum $\mathbf{H_G}$, which is constant since $\sum M_G = 0$

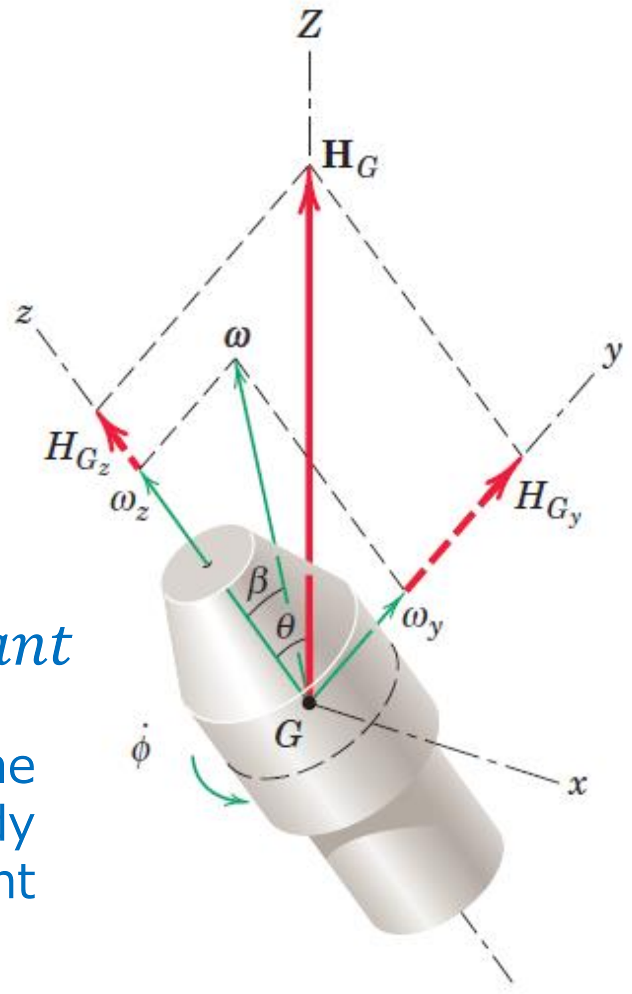
$$H_{G_x} = 0$$
$$H_{G_y} = H_G \sin\theta$$
$$H_{G_z} = H_G \cos\theta$$

$$H_{G_x} = 0 = I_o\omega_x$$
$$H_{G_y} = H_G \sin\theta = I_o\omega_y$$
$$H_{G_z} = H_G \cos\theta = I\omega_z$$

$$H_x = [(I_{xx})\omega_x - I_{xy}\omega_y - I_{xz}\omega_z]$$
$$H_y = [-I_{xy}\omega_x + (I_{yy})\omega_y - I_{yz}\omega_z]$$
$$H_z = [-I_{zx}\omega_x - I_{zy}\omega_y + (I_{zz})\omega_z]$$

$$\omega_x = \Omega_x = 0 \qquad \theta = constant$$

This result means that the motion is one of steady precession about the constant $\mathbf{H_G}$ vector.

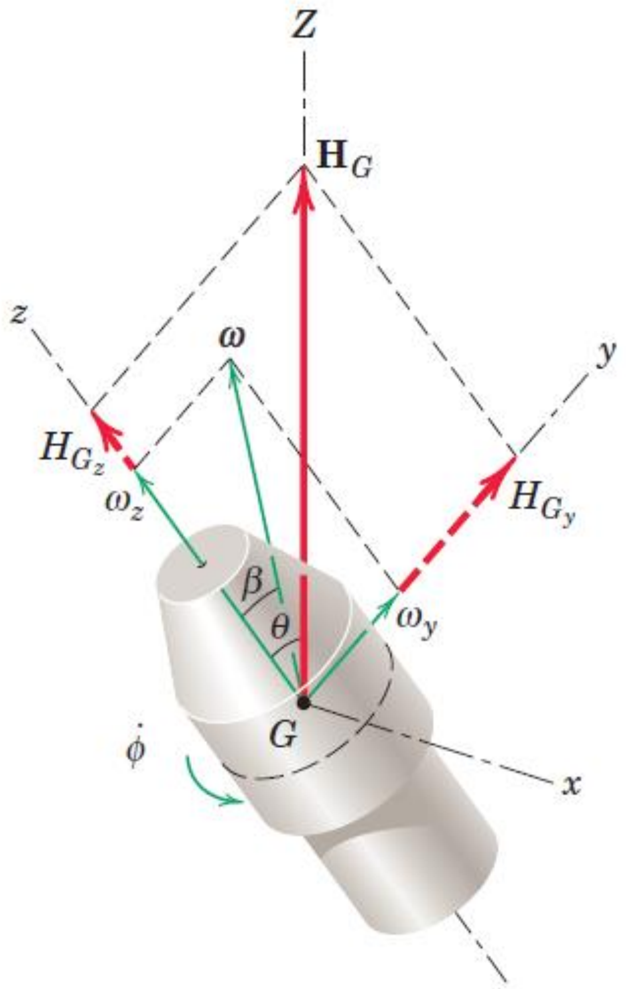


Steady State Precession with Zero Moment

With no x-component, the angular velocity $\boldsymbol{\omega}$ of the rotor lies in the y-z plane along with the Z-axis and makes an angle β with the z-axis. The relationship between β and θ is obtained from

$$\tan\theta = \frac{H_{G_y}}{H_{G_z}} = \frac{I_o \omega_y}{I \omega_z} = \frac{I_o}{I} \tan\beta$$

Angular velocity $\boldsymbol{\omega}$ makes a constant angle β with the spin axis



Steady State Precession with Zero Moment

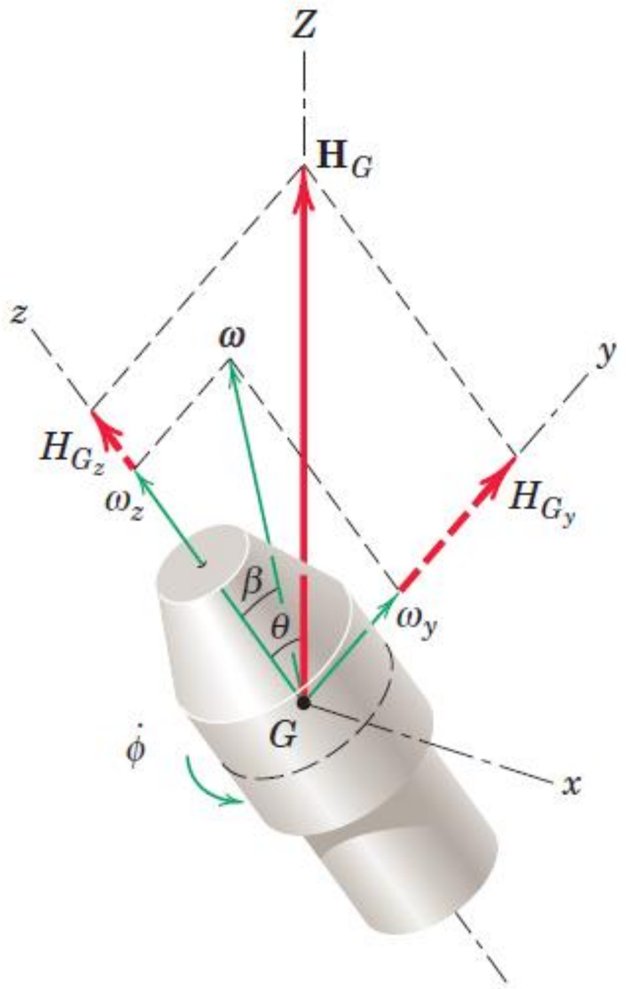
The rate of precession is obtained from

$$\sum M_x = \dot{\psi}\sin\theta\,\big(\dot{\psi}\ \cos\theta\big)(I - I_o) + I\dot{\psi}\,p\sin\theta = 0$$

$$\big(\dot{\psi}\ \cos\theta\big)(I - I_o) + I\,p = 0$$

$$\dot{\psi} = \frac{I\,p}{(I_o - I)\cos\theta}$$

Direction of the precession depends on the relative magnitudes of the two moments of inertia



Steady State Precession with Zero Moment

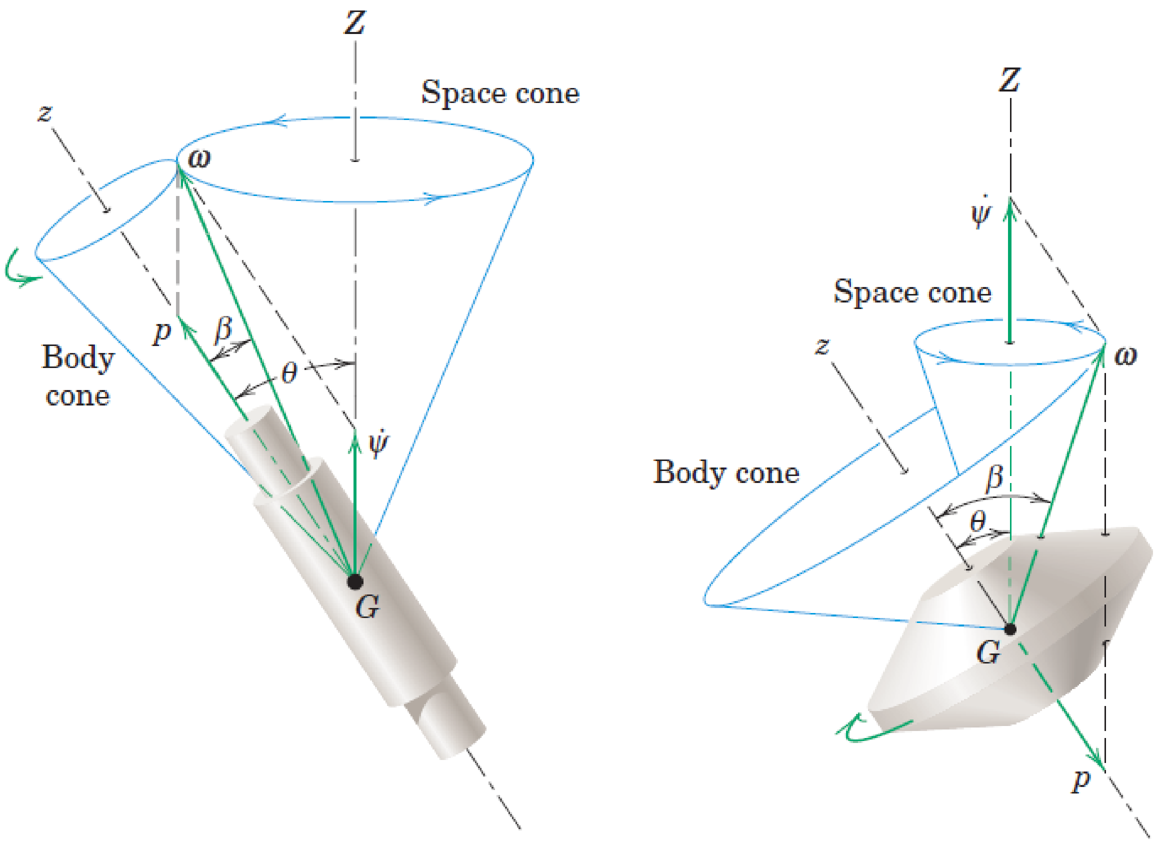
The rate of precession is obtained from

$$\dot{\psi} = \frac{I\,p}{(I_o - I)\cos\theta} \qquad \tan\theta = \frac{I_o}{I}\tan\beta$$

Direction of the precession depends on the relative magnitudes of the two moments of inertia

If $I_0 > I$, then $\beta < \theta$, and the precession is said to be ***direct***.

The body cone rolls on the outside of the space cone.



Direct precession $I_0 > I$

(a)

Retrograde precession $I_0 < I$

(b)

Steady State Precession with Zero Moment

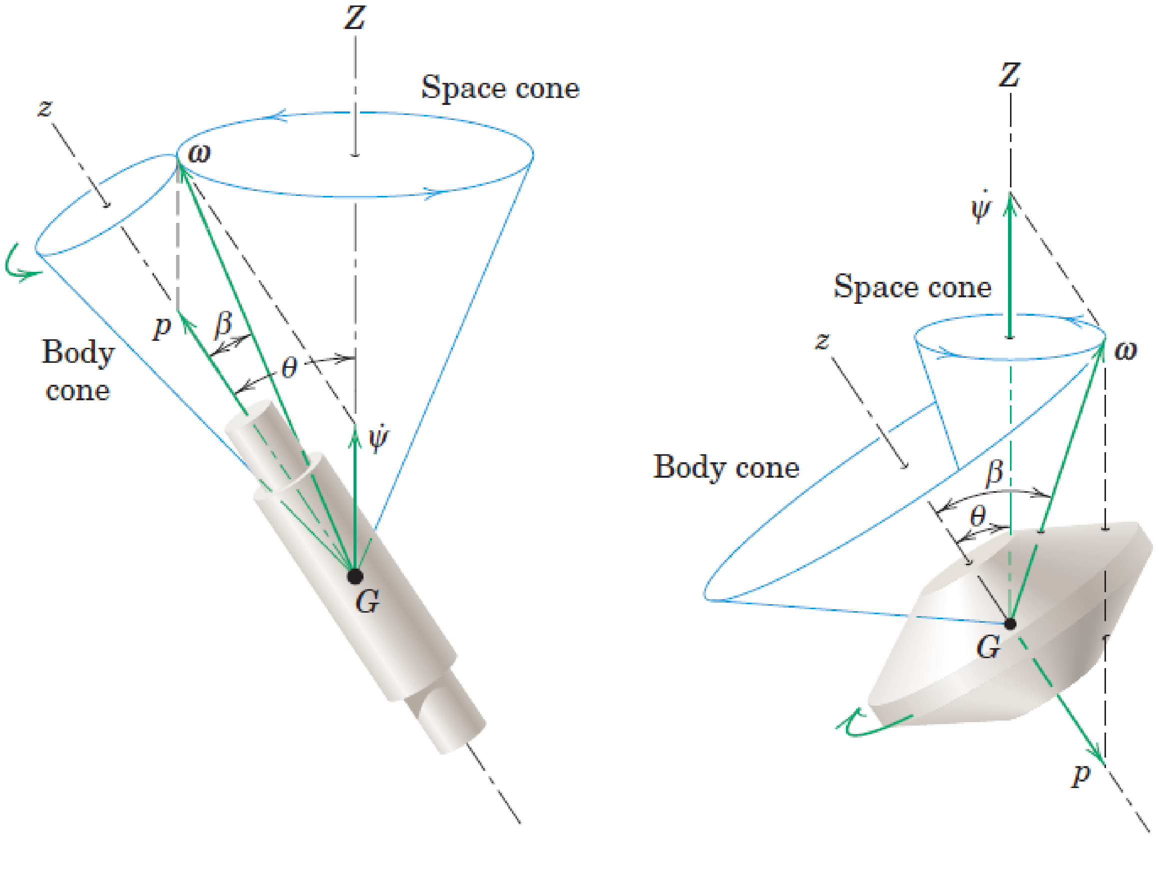
The rate of precession is easily obtained from

$$\dot{\psi} = \frac{I\,p}{(I_o - I)\cos\theta} \qquad \tan\theta = \frac{I_o}{I}\tan\beta$$

Direction of the precession depends on the relative magnitudes of the two moments of inertia

If $I_0 < I$, then $\beta > \theta$, and the precession is said to be ***retrograde***.

Space cone is internal to the body cone, and $\dot{\psi}$ and p have opposite signs



Direct precession $I_0 > I$

(a)

Retrograde precession $I_0 < I$

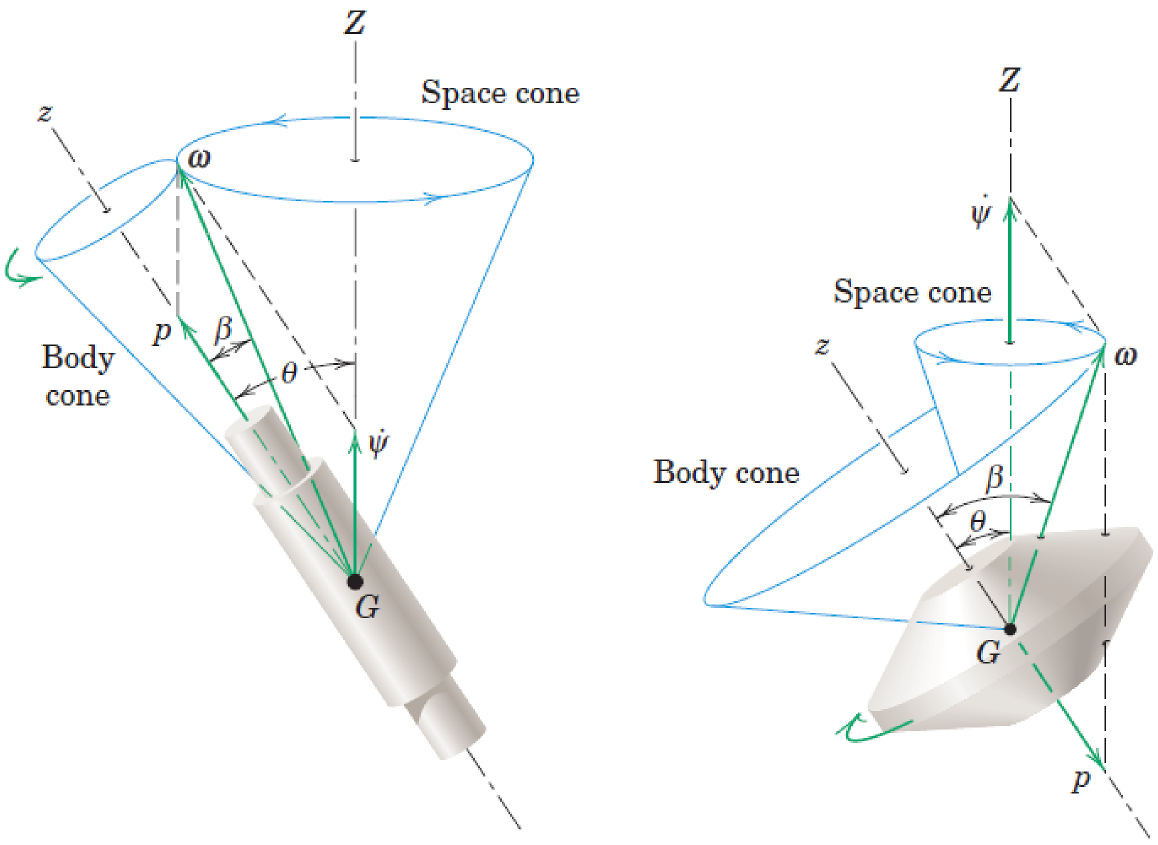
(b)

Steady State Precession with Zero Moment

The rate of precession is easily obtained from

$$\tan\theta = \frac{I_o}{I}\tan\beta$$

- If $I = I_0$, above equation and figure show that both angles must be zero to be equal.
- For this case, the body has no precession and merely rotates with an angular velocity **p**.
- This condition occurs for a body with point symmetry, such as with a homogeneous sphere.



Direct precession $I_0 > I$

(a)

Retrograde precession $I_0 < I$

(b)

THANK YOU